

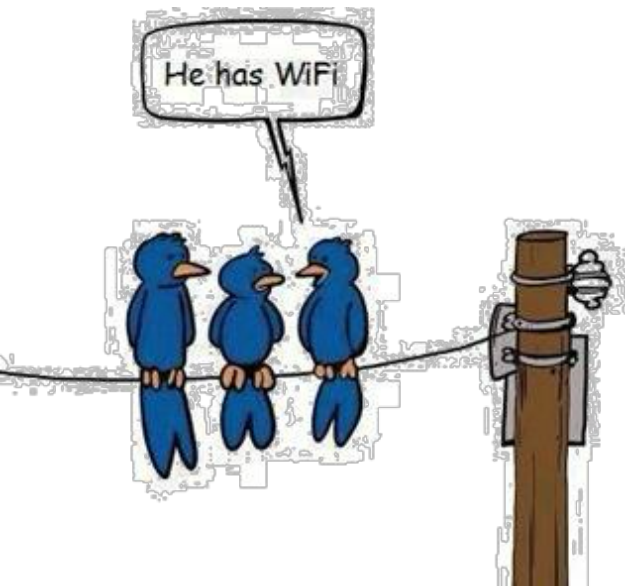
ECSE 4560/ 6560

Modern Communication Systems

Lecture 22: Wireless Sensing Part 3:

AOA-TOF

Ish Jain, Fall 2025



Announcements

- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **AOA-TOF**
- Projects
 - Group formation [MCS_Fall25_project.xlsx](#)
 - Final Project Presentation and Demos: Dec 9 and Dec 12
 - Project Report and peer evaluation: Finals exam week
- HW/Exams Dates
 - HW6 (MIMO code) due **Mon 11/17**
 - HW7 (Sensing Theory) due **Fri 11/21** – covered in Exam 2
 - Exam 2: **Friday Nov 21** - Anyone needs makeup exam?

60-75% MIMO -

30-40%
sensing

Extra Reading on Piazza

- Principles of Modern Wireless Communication Systems Theory and Practice; by Aditya K. Jagannatham, IIT Kanpur Book
 - https://piazza.com/class_profile/get_resource/mevvk9kirhg2fu/mgb9esiush64ae
 - Parts of Chapter 1,2,3,4,7
 - **MIMO Chapter 6**
 - Sensing Readings
 - Xiong, Jie, and Kyle Jamieson. "ArrayTrack: A Fine-Grained indoor location system." *NSDI 2013*
 - 2024 SIGMOBILE Test-of-Time Award 
- spot fi - ACM sigcomm'15 (Tof) with ↓

Today's Class

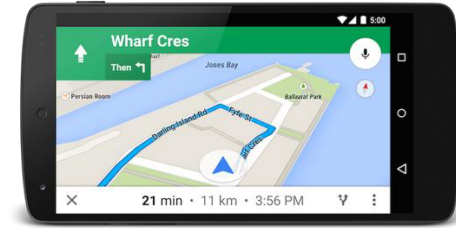
- **Wireless Sensing**
 - **Localization Techniques**
 - **AOA-TOF based localization**

Wireless Localization / Positioning

The process of obtaining a human or object's location using wireless signals

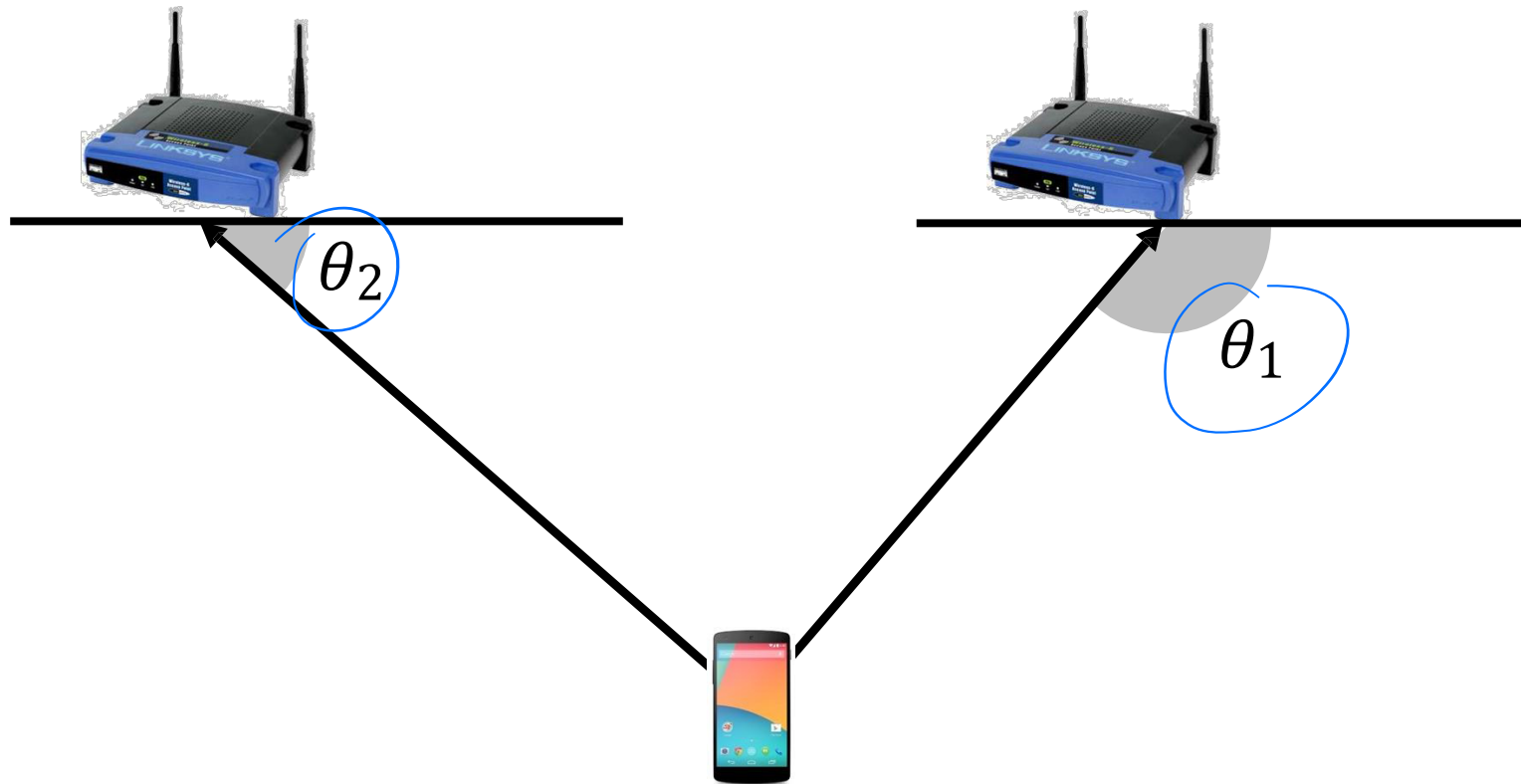
Applications:

- Navigation: outdoors (GPS) and indoors (e.g., museum)
- Location based services: Tagging, Reminder, Ads
- Virtual Reality and Motion Capture
- Gestures, writing in the air
- Behavioral Analytics (Health, activities, etc.)
- Locating misplaced items (keys)
- Location based security
- Delivery drones



Method 2: AoA Based Localization

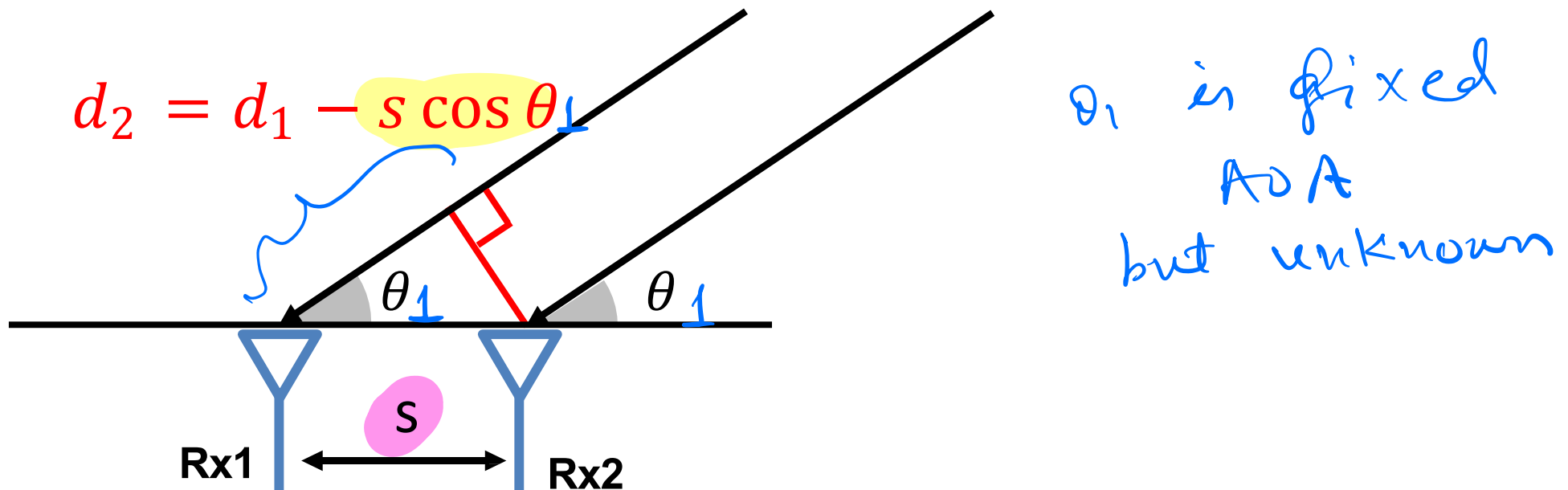
Measure Angle of Arrival (AoA) from device to each AP



Triangulation

Method 2: AoA Based Localization

Measure Angle of Arrival (AoA) from device to each AP

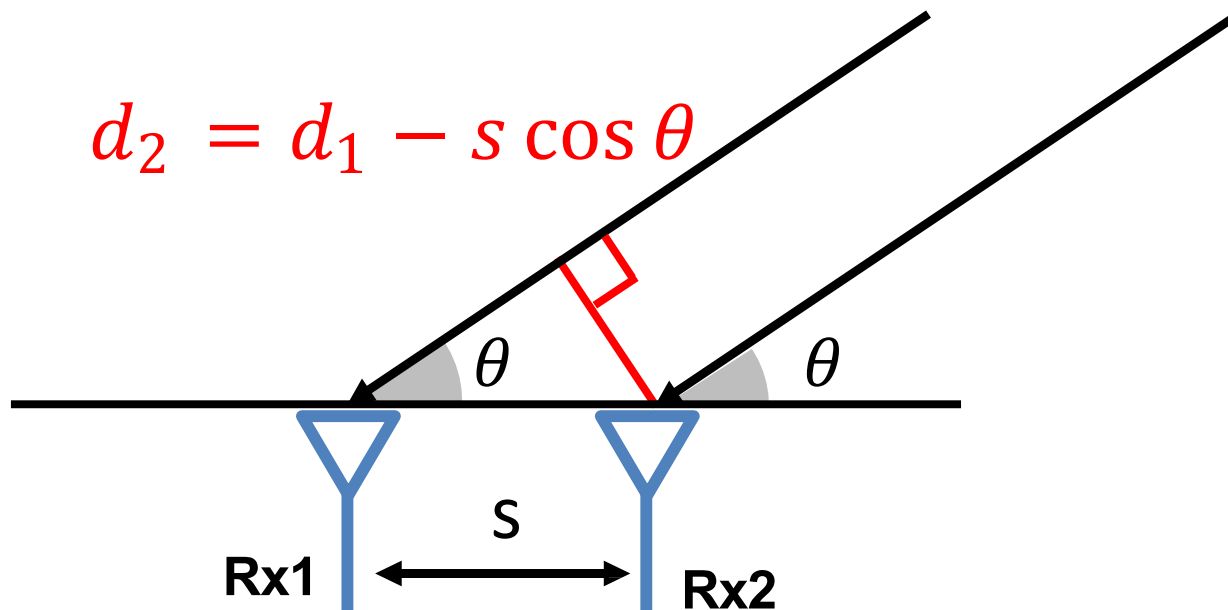


$$h_1 \propto e^{-j2\pi \frac{d_1}{\lambda}} \quad h_2 \propto e^{-j2\pi \frac{d_2}{\lambda}} = e^{-j2\pi \frac{d_1 - s \cos \theta_1}{\lambda}}$$

$$\Delta\Phi = \angle h_2 - \angle h_1 = 2\pi s \cos \theta_1 / \lambda \bmod 2\pi$$

Method 2: AoA Based Localization

Measure Angle of Arrival (AoA) from device to each AP



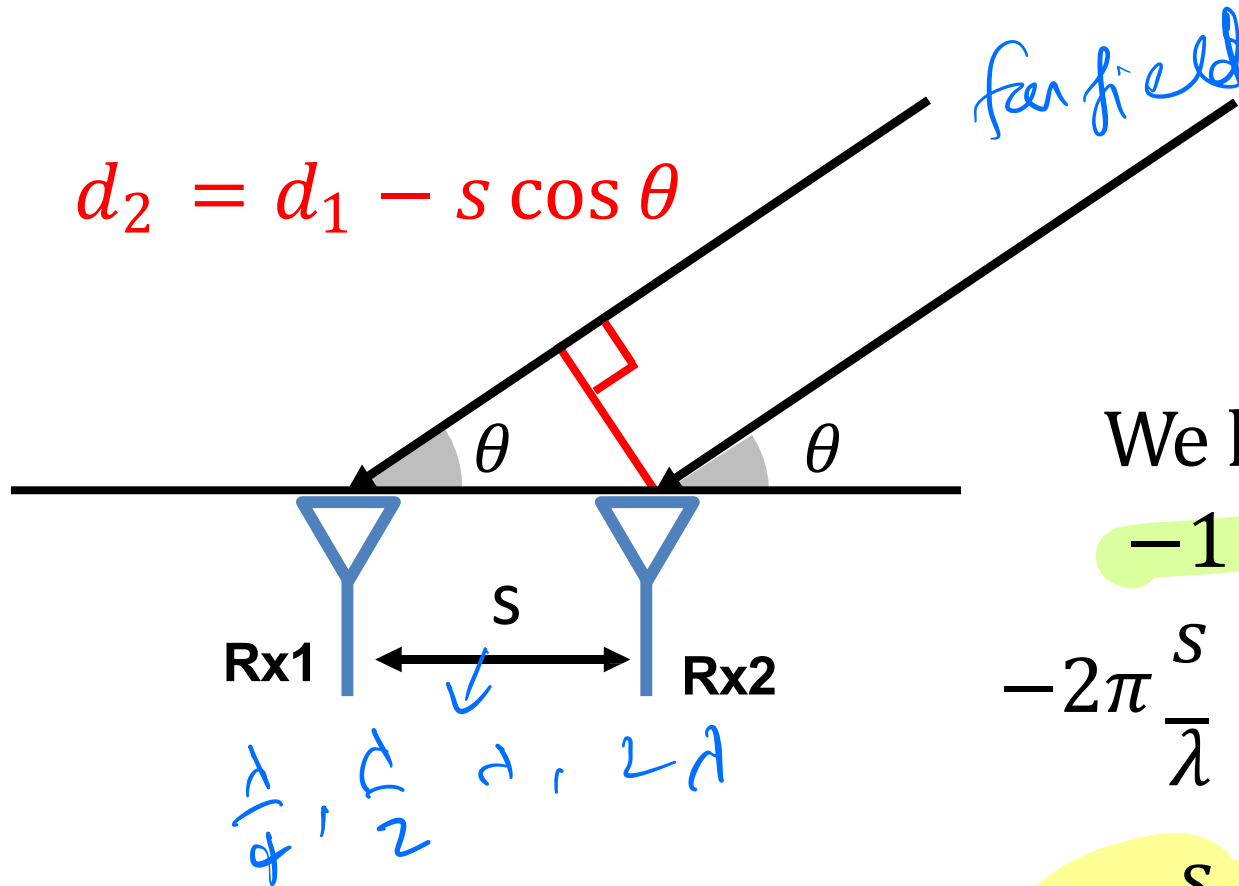
$$\Delta\Phi = \angle h_2 - \angle h_1 = 2\pi s \cos \theta / \lambda \mod 2\pi$$

Ambiguity: $\exists \theta_1 \neq \theta_2 \mid \Delta\Phi_1 = \Delta\Phi_2 \mod 2\pi$

To avoid ambiguity, we want: $-\pi \leq \Delta\Phi \leq \pi$

Method 2: AoA Based Localization

Measure Angle of Arrival (AoA) from device to each AP



We want:

$$-\pi \leq \Delta\Phi \leq \pi$$

We have:

$$-1 \leq \cos \theta \leq 1$$

$$-2\pi \frac{s}{\lambda} \leq 2\pi \frac{s}{\lambda} \cos \theta \leq 2\pi \frac{s}{\lambda}$$

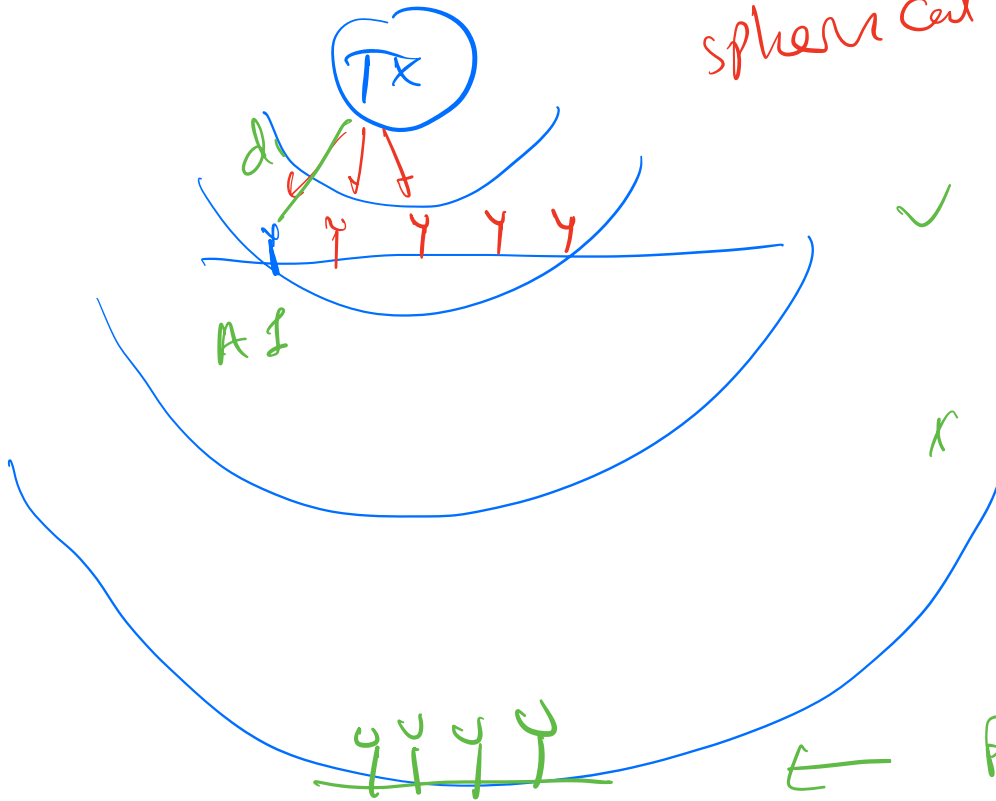
$$-2\pi \frac{s}{\lambda} \leq \Delta\Phi \leq 2\pi \frac{s}{\lambda}$$

Set: $s = \lambda/2$

$$-\pi \leq \Delta\Phi \leq \pi$$

what if near-field

spherical wave in
nearfield



$$h \propto \frac{1}{d_1} e^{-j2\pi \frac{d_1}{\lambda}}$$

$$d_2 = d_1 \cos \theta$$

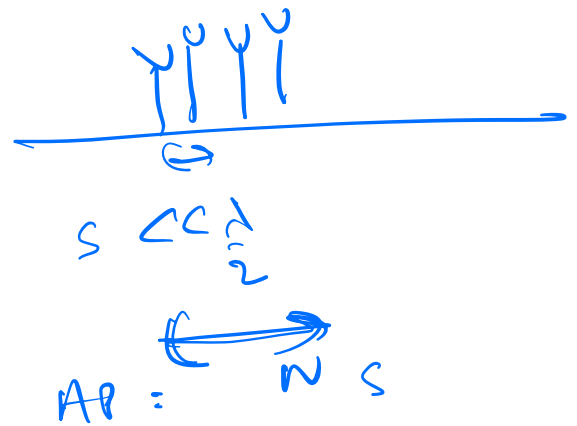
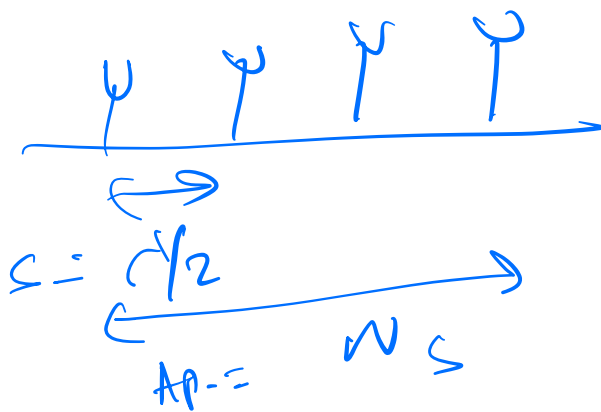
planar wave
approxim.
in far-field

is $s \leq \frac{\lambda}{2} \Rightarrow$ no ambiguity

$s > \frac{\lambda}{2} \Rightarrow$ ambiguity

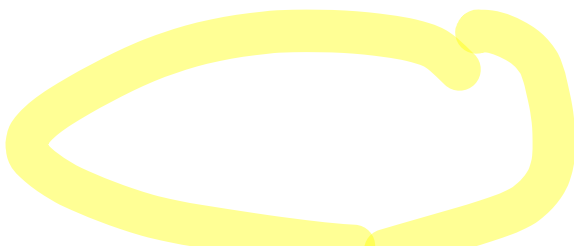
But $s \ll \frac{\lambda}{2} \Rightarrow$ antenna coupling

$\Rightarrow$ small aperture



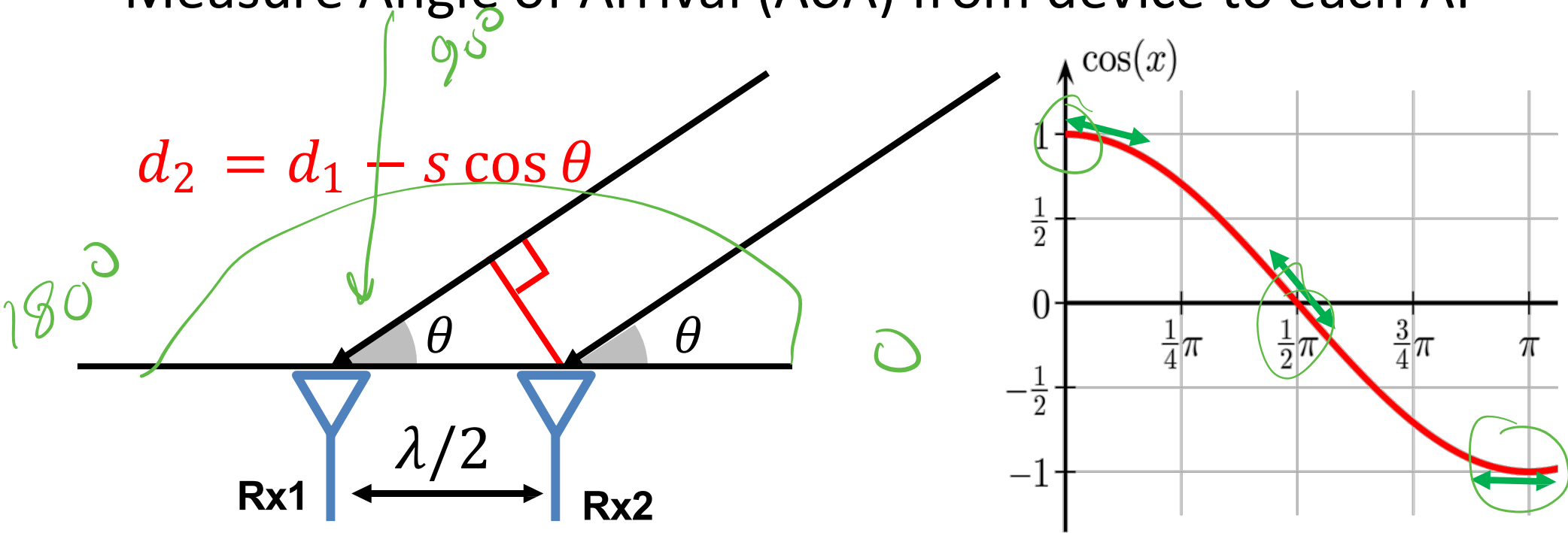
low aperture

$\Rightarrow$ low SNR



Method 2: AoA Based Localization

Measure Angle of Arrival (AoA) from device to each AP



Pros: More accurate than RSSI, Simple!

Cons: Ambiguity: $\cos \theta = \cos(-\theta)$

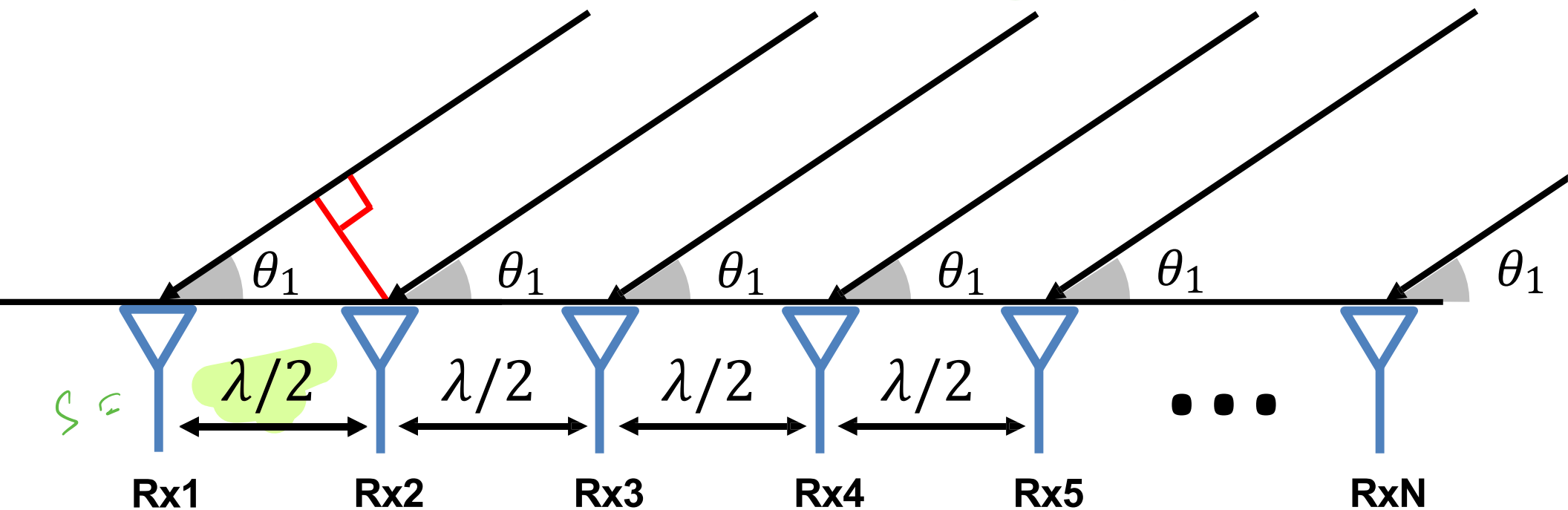
Error not linear with θ due to $\cos \theta$

Requires 2 Antennas separated $\lambda/2$

Does not work with multipath!

irrespective of antenna spacing

Method 3: Antenna Arrays (ArrayTrack Paper)



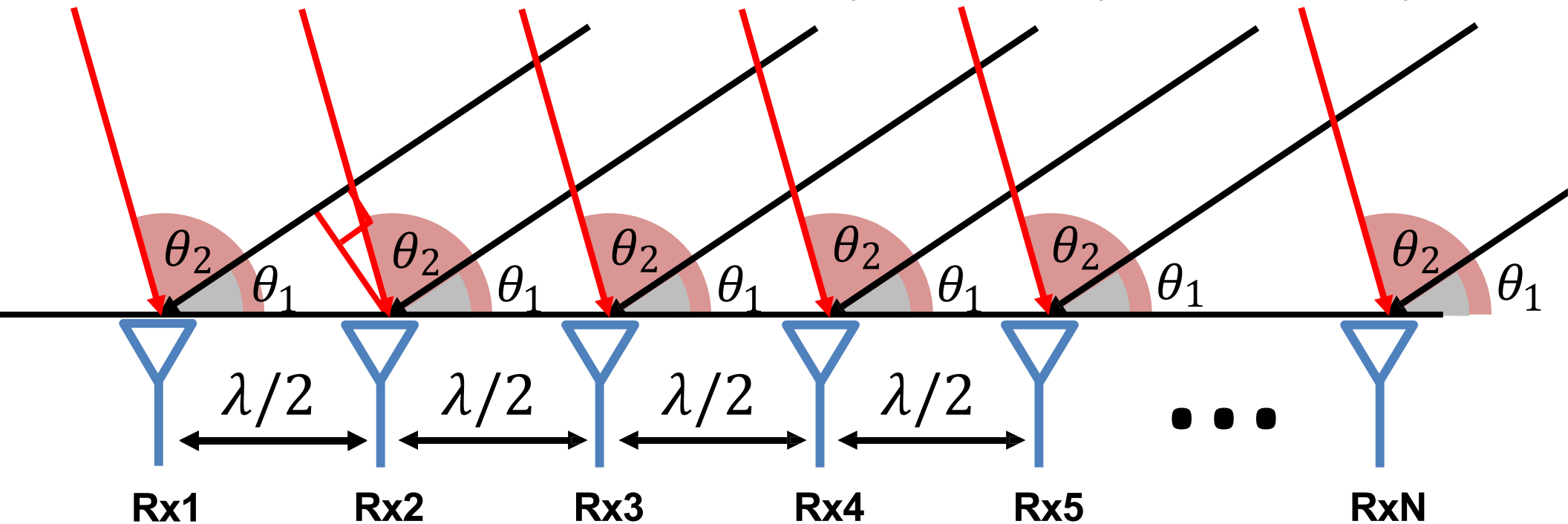
$$h_k = \alpha_1 e^{-j2\pi \frac{d_1 - k s \cos \theta_1}{\lambda}} = \alpha_1 e^{-j\phi_1 + k\pi \cos \theta_1}$$

$$\angle h_2 - \angle h_1 = \pi \cos \theta_1$$

$$\angle h_{k+1} - \angle h_k = \pi \cos \theta_1$$

$$\angle h_k - \angle h_1 = k\pi \cos \theta_1$$

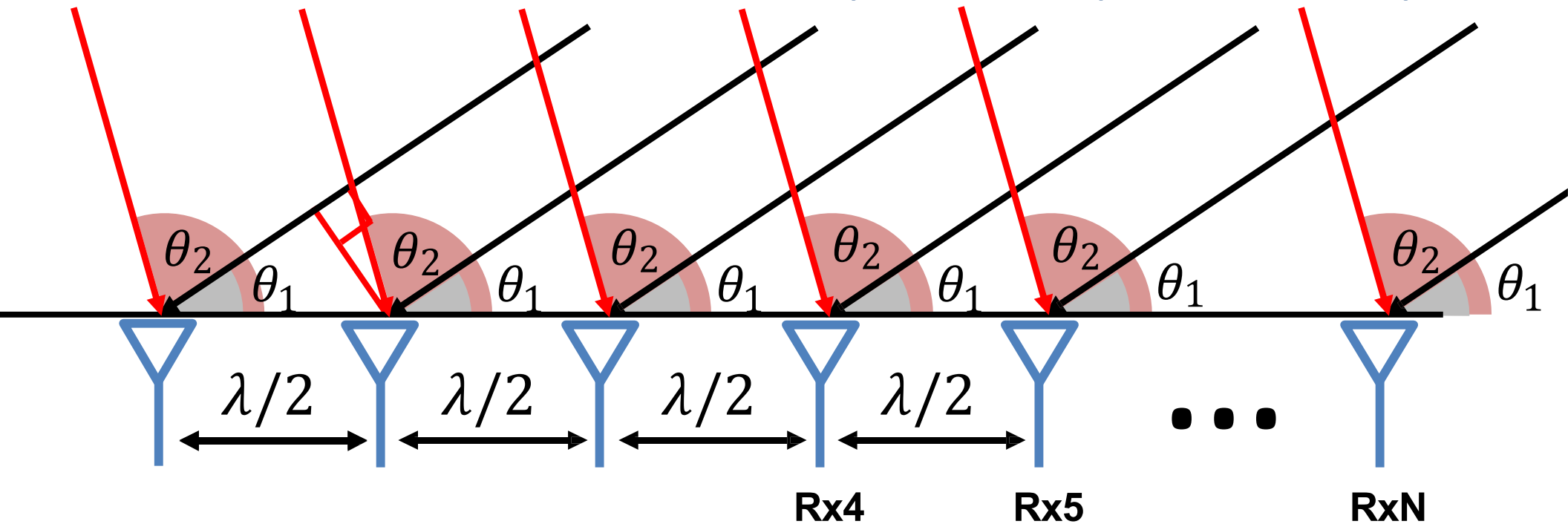
Method 3: Antenna Arrays (ArrayTrack Paper)



$$h_k = \alpha_1 e^{-j2\pi \frac{d_1 - k s \cos \theta_1}{\lambda}} = \alpha_1 e^{-j\phi_1 + k\pi \cos \theta_1} \\ + \alpha_2 e^{-j2\pi \frac{d_2 - k s \cos \theta_2}{\lambda}} + \alpha_2 e^{-j\phi_2 + k\pi \cos \theta_2}$$

For L paths $\rightarrow$
$$h_k = \sum_{l=1}^L \alpha_l e^{jk\pi \cos \theta_l - j\phi_l}$$

Method 3: Antenna Arrays (ArrayTrack Paper)



$$h_k = \sum_{l=1}^L \alpha_l e^{jk\pi \cos \theta_l - j\phi_l}$$

0 to 180°
↓

Multipath Profile: $P(\theta)$
(Beam pattern)

$$P(\theta) = \left| \sum_{k=1}^N h_k e^{-jk\pi \cos \theta} \right|^2$$

↑ time ↑ freq

Similar to FFT
It's called spatial FFT

Method 3: Antenna Arrays (ArrayTrack Paper)

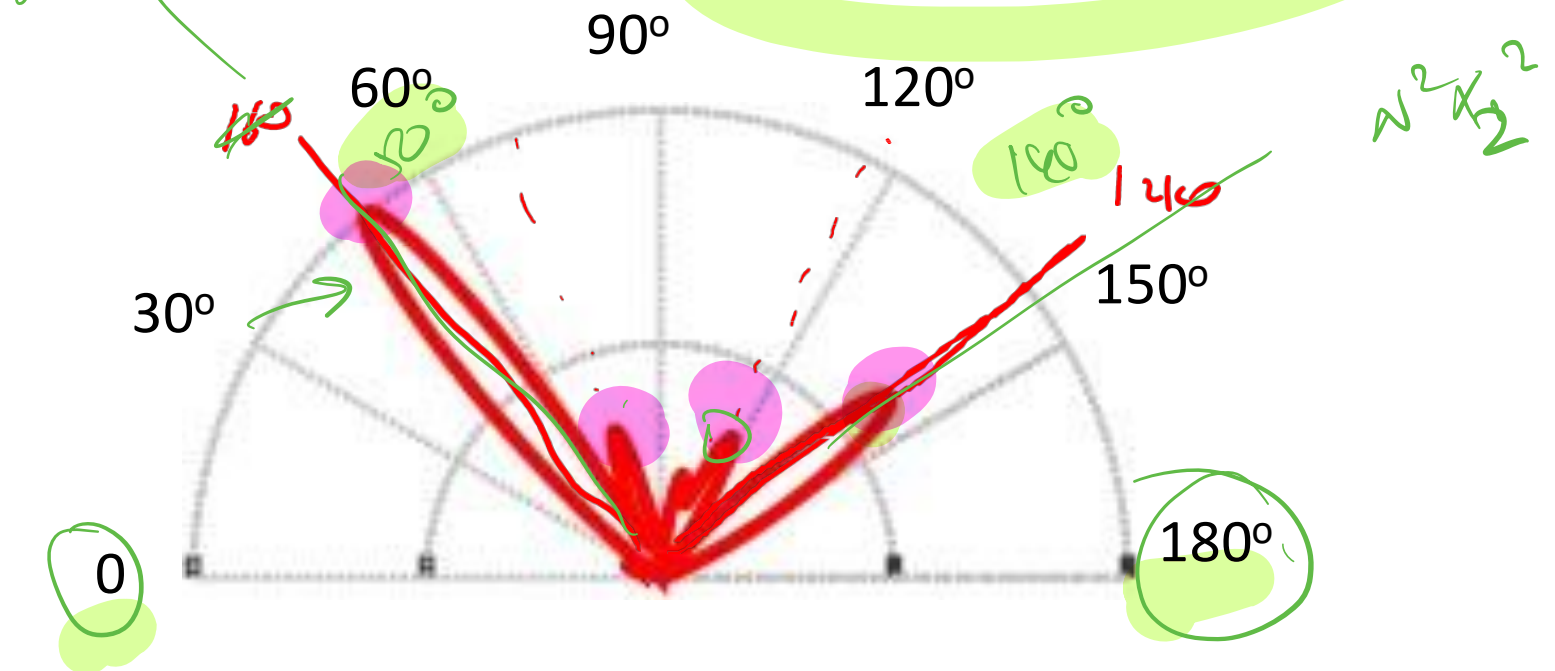
$$\begin{aligned}
 P(\theta) &= \left| \sum_{k=1}^N h_k e^{-jk\pi \cos \theta} \right|^2 \\
 P(\theta_1) &= \left| \sum_{k=1}^N h_k e^{-jk\pi \cos \theta_1} \right|^2 \\
 &= \left| \sum_{k=1}^N \left(\sum_{l=1}^L \alpha_l e^{jk\pi \cos \theta_l - j\phi_l} \right) e^{-jk\pi \cos \theta_1} \right|^2 \\
 &= \left| \sum_{k=1}^N \alpha_1 e^{-j\phi_1} + \sum_{k=1}^N \left(\sum_{l=2}^L \alpha_l e^{jk\pi (\cos \theta_l - \cos \theta_1) - j\phi_l} \right) \right|^2 \\
 &= \left| N\alpha_1 e^{-j\phi_1} + \sum_{l=2}^L \alpha_l e^{-j\phi_l} \left(\sum_{k=1}^N e^{jk\pi (\cos \theta_l - \cos \theta_1)} \right) \right|^2 \approx N^2 \alpha_1^2
 \end{aligned}$$

$\theta = \theta_1$
 α_l is channel magnitude for path l
 introduce sidelobes

Method 3: Antenna Arrays (ArrayTrack Paper)

$$h_k = \sum_{l=1}^L \alpha_l e^{jk\pi \cos \theta_l - j\phi_l}$$

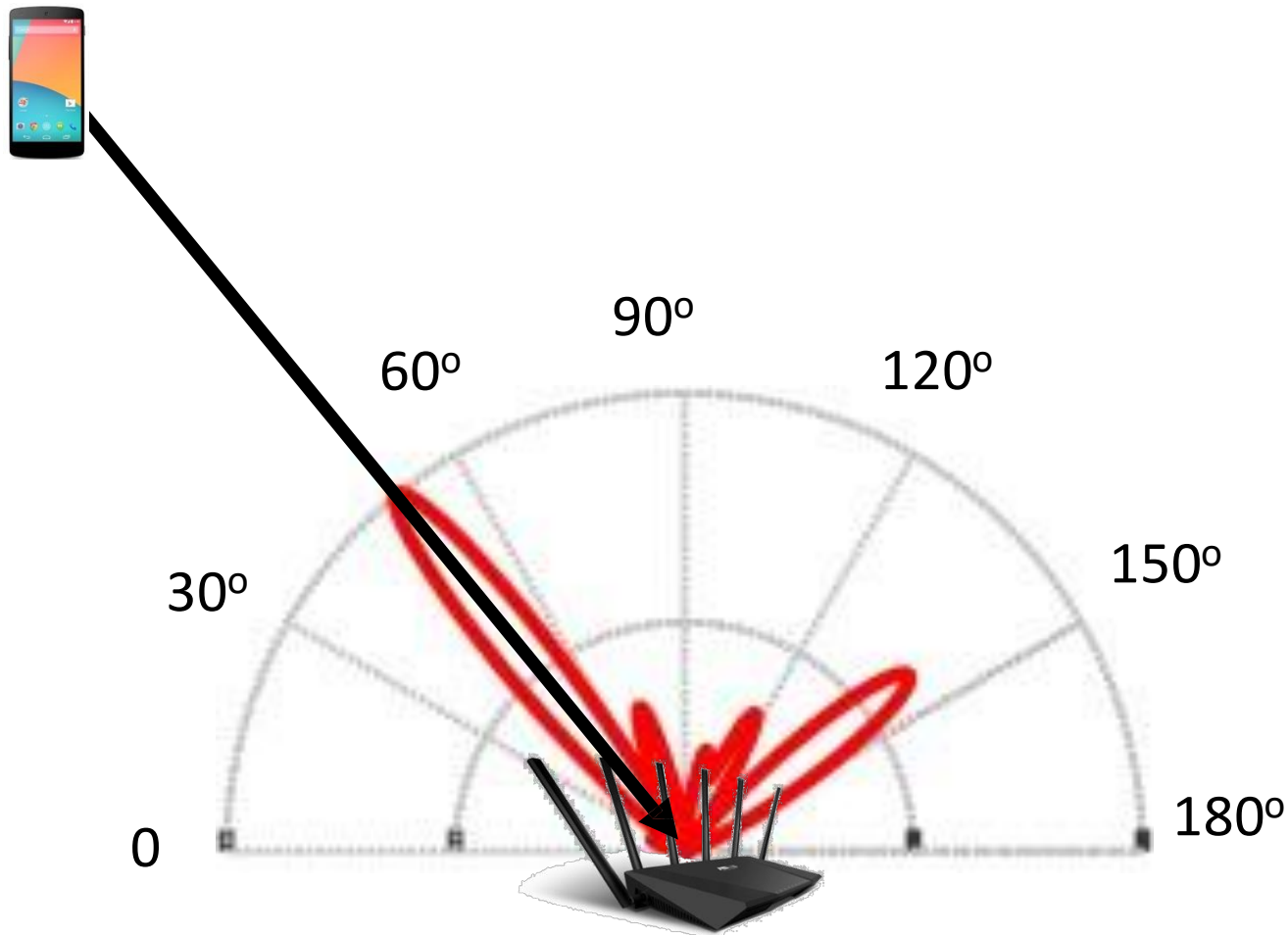
Multipath Profile: $P(\theta) = \left| \sum_{k=1}^N h_k e^{-jk\pi \cos \theta} \right|^2$



Method 3: Antenna Arrays (ArrayTrack Paper)

Which is the Line-of-Sight Path (Direct Path)?

Strongest Path!

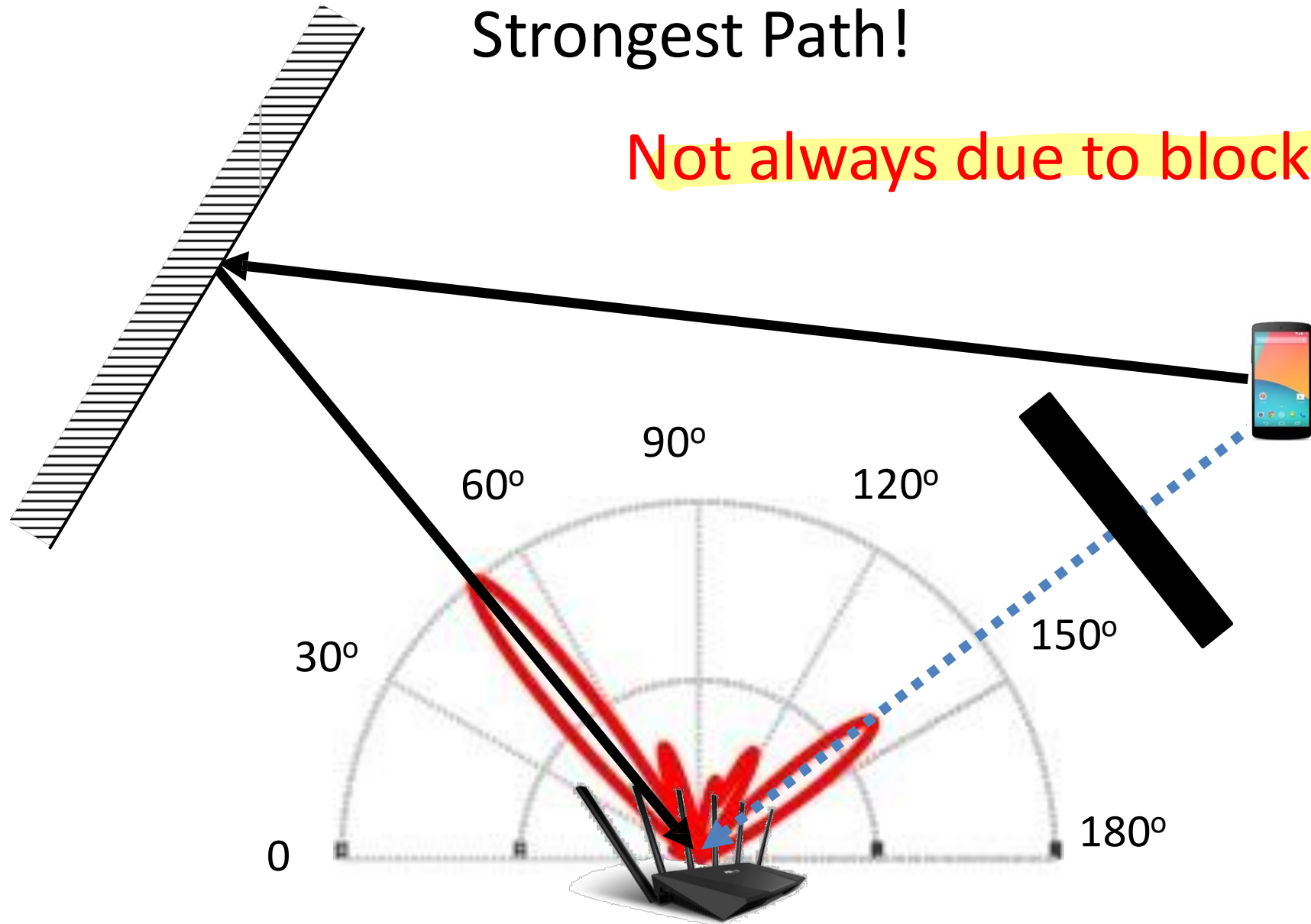


Method 3: Antenna Arrays (ArrayTrack Paper)

Which is the Line-of-Sight Path (Direct Path)?

Strongest Path!

Not always due to blockage!

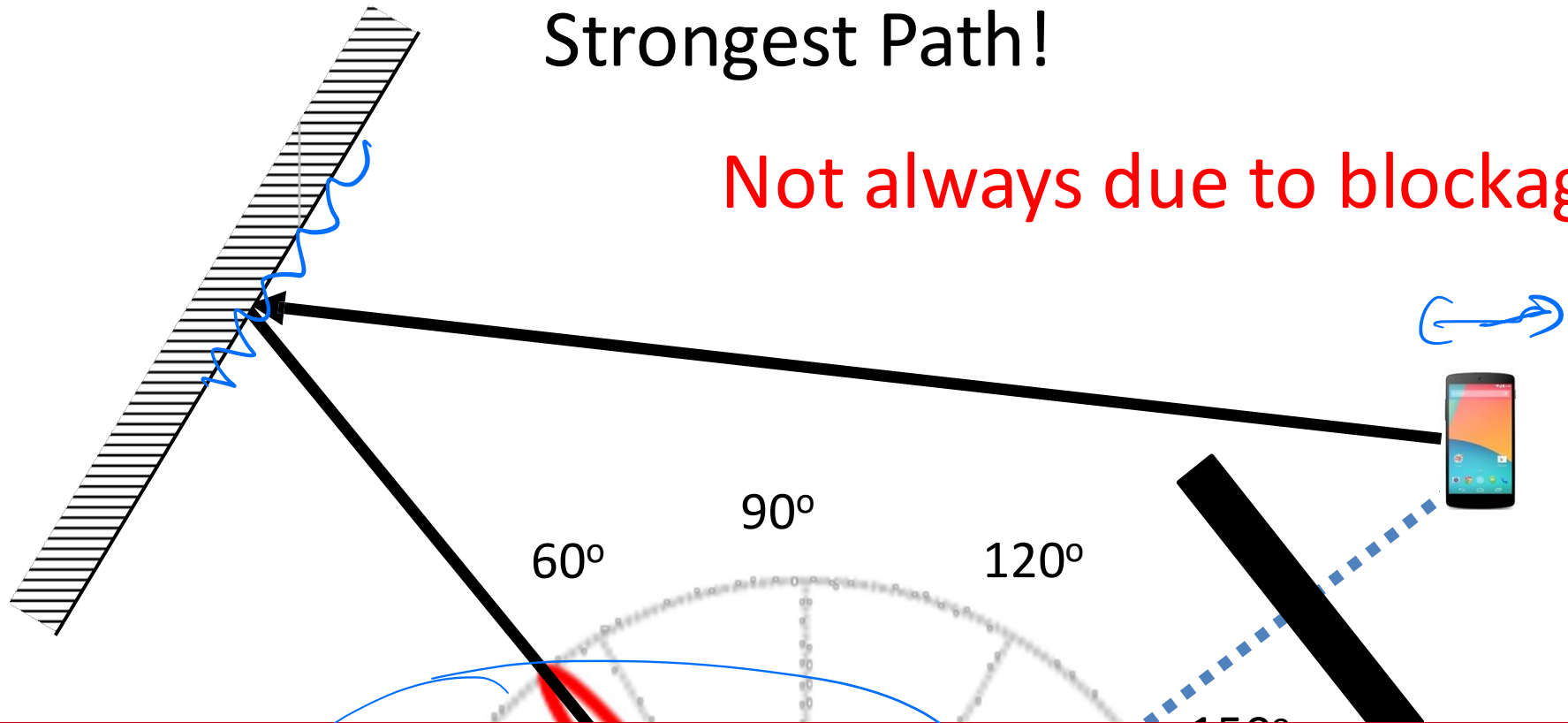


Method 3: Antenna Arrays (ArrayTrack Paper)

Which is the Line-of-Sight Path (Direct Path)?

Strongest Path!

Not always due to blockage!



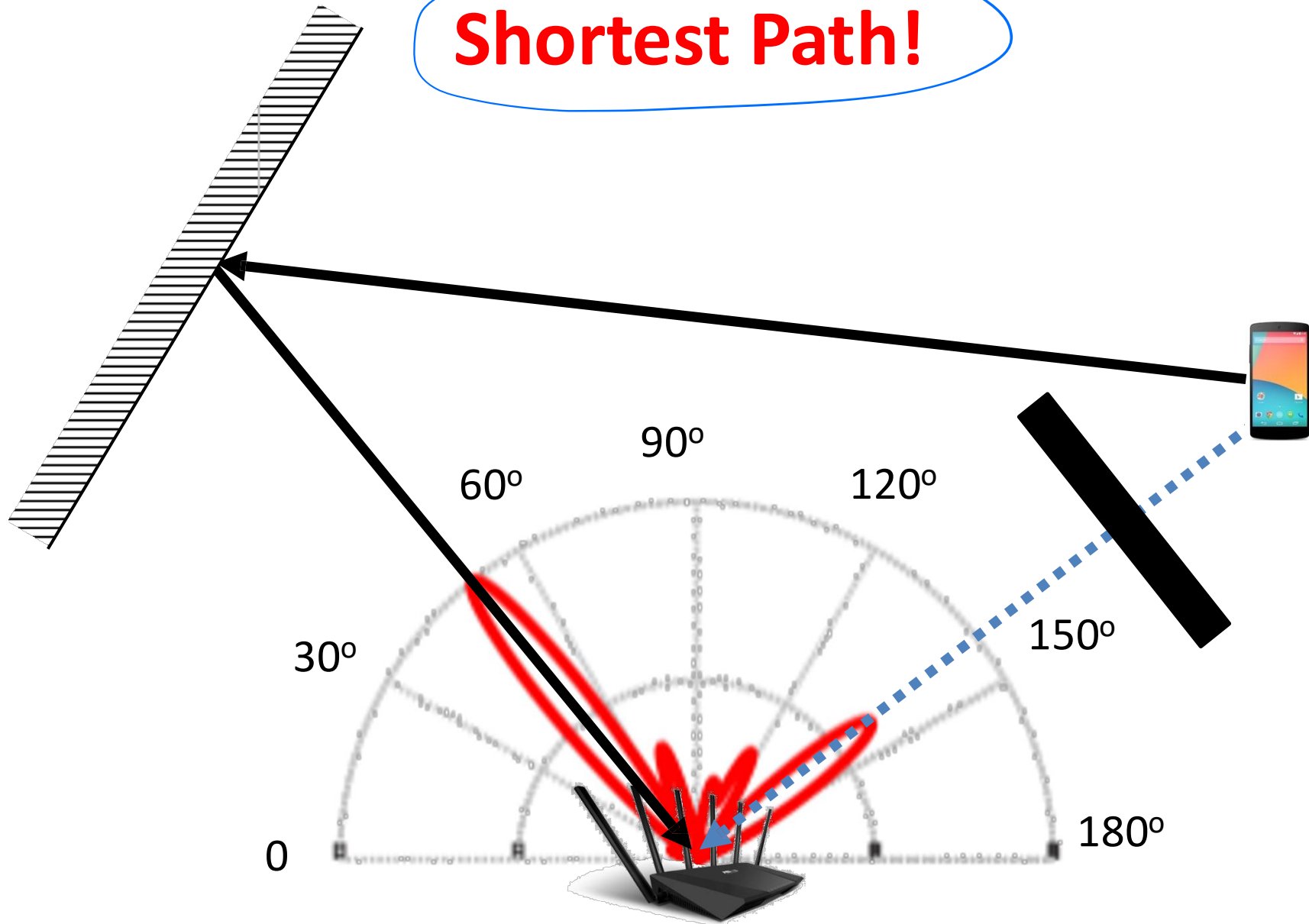
ArrayTrack: Leverage Mobility

- Line of sight path relatively stable with mobility
- Multipath reflection changes faster with mobility

Method 3: Antenna Arrays

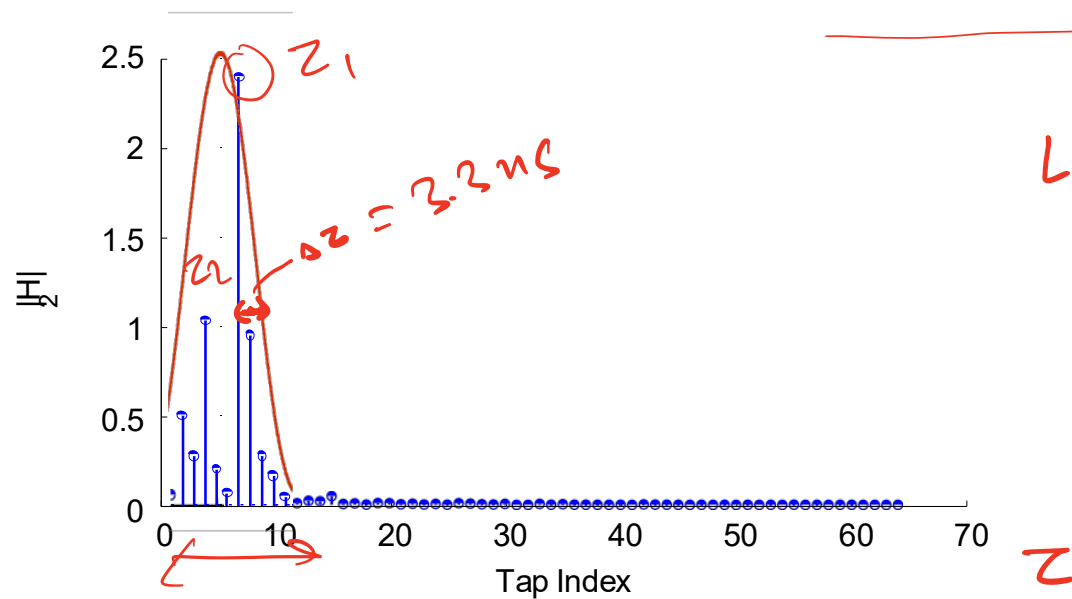
Which is the Line-of-Sight Path (Direct Path)?

Shortest Path!



Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?



OFDM $\rightarrow H(f) \xrightarrow{\text{IFFT}} h(z)$
 L paths $\Rightarrow$ how to separate them

Multipath Profile vs Time

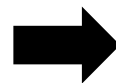
resolution $\Rightarrow$ two path separated by Δz

$$\Delta \tau = \frac{\Delta d}{c}$$

$$\Delta d = 1 \text{ m}$$

$$c = 3 \times 10^8 \text{ m/s}$$

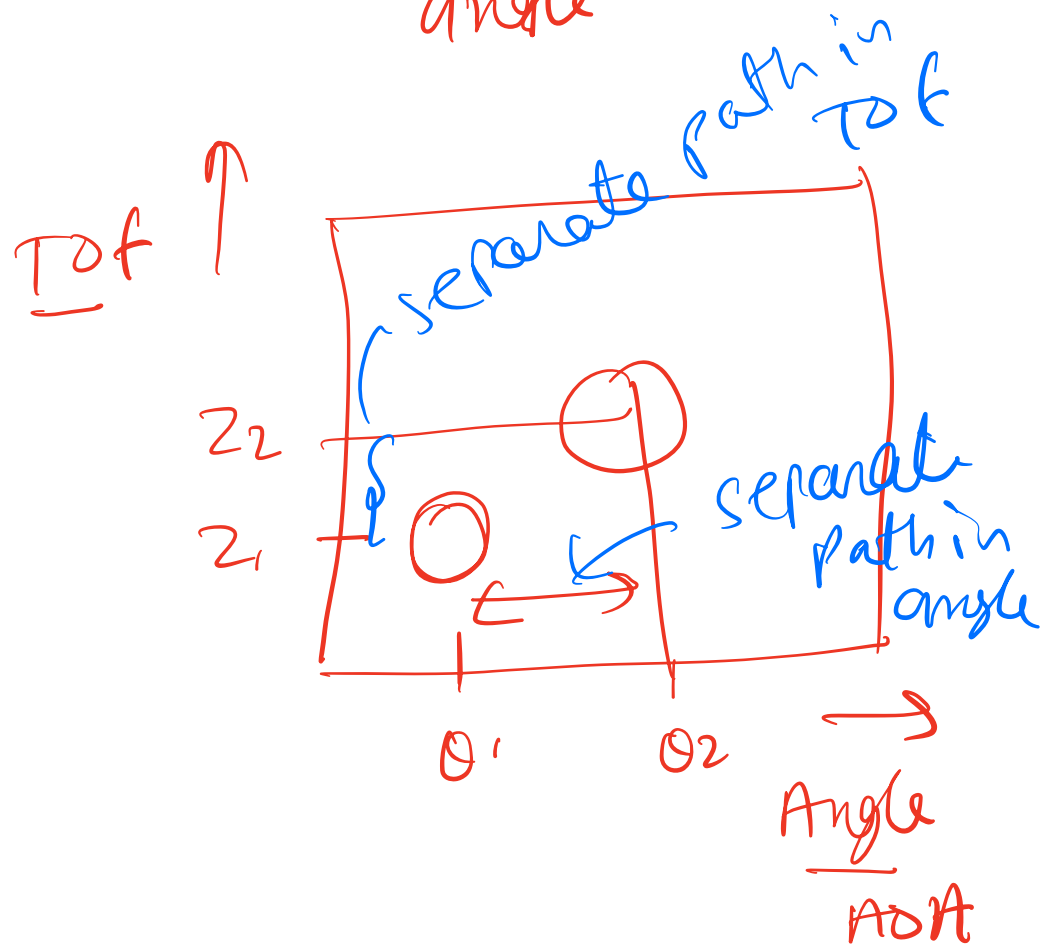
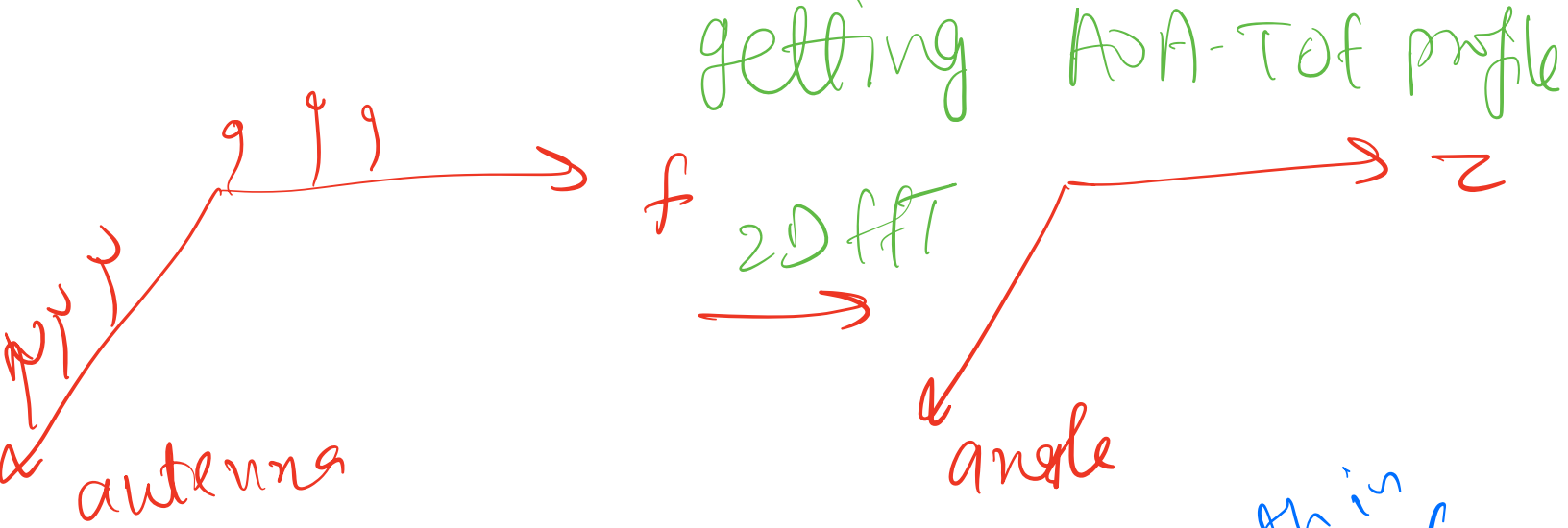
$$\Delta \tau = 3.3 \text{ ns}$$

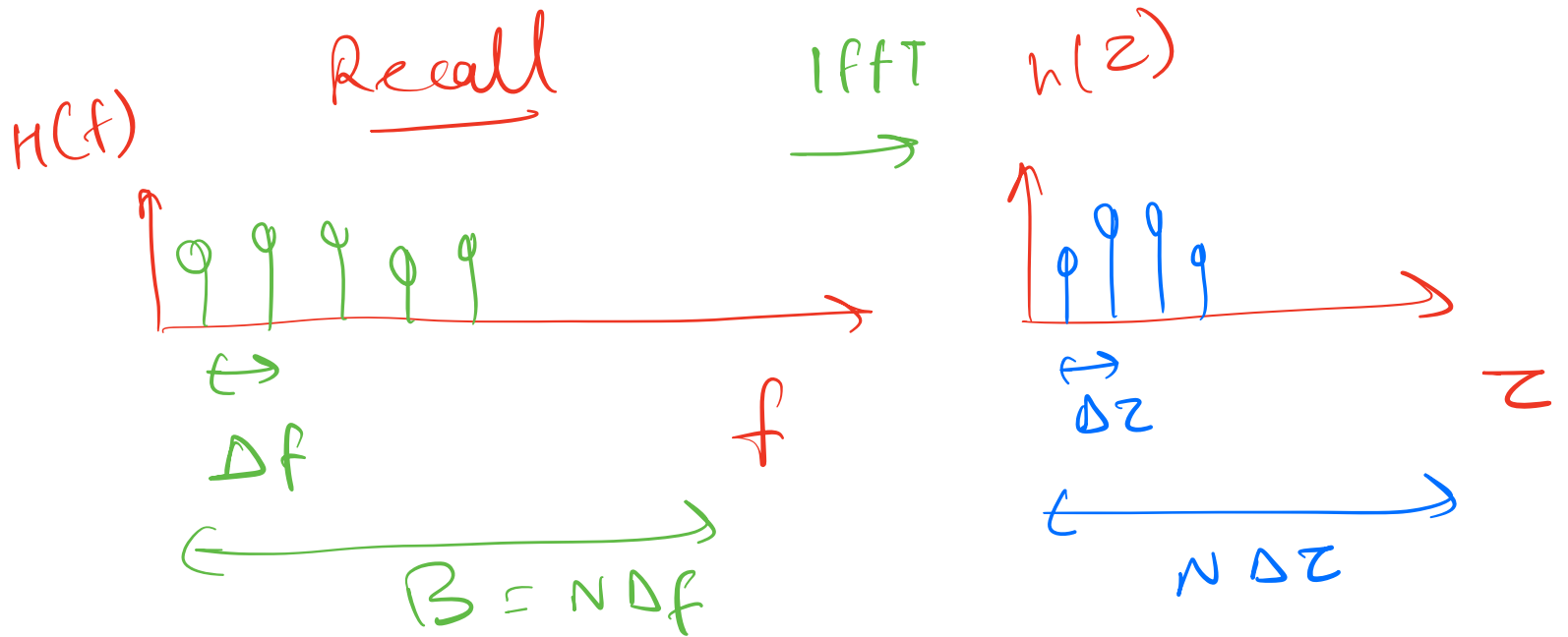


Requires a sampling rate

$$1/\Delta \tau = \underline{300 \text{ MHz}}$$

802.11n bandwidth = 40 MHz

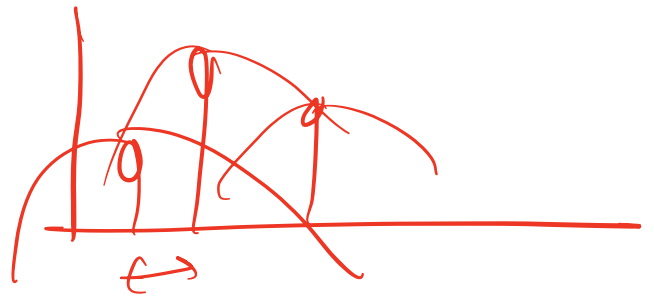




$$\Delta z = \frac{1}{B}$$

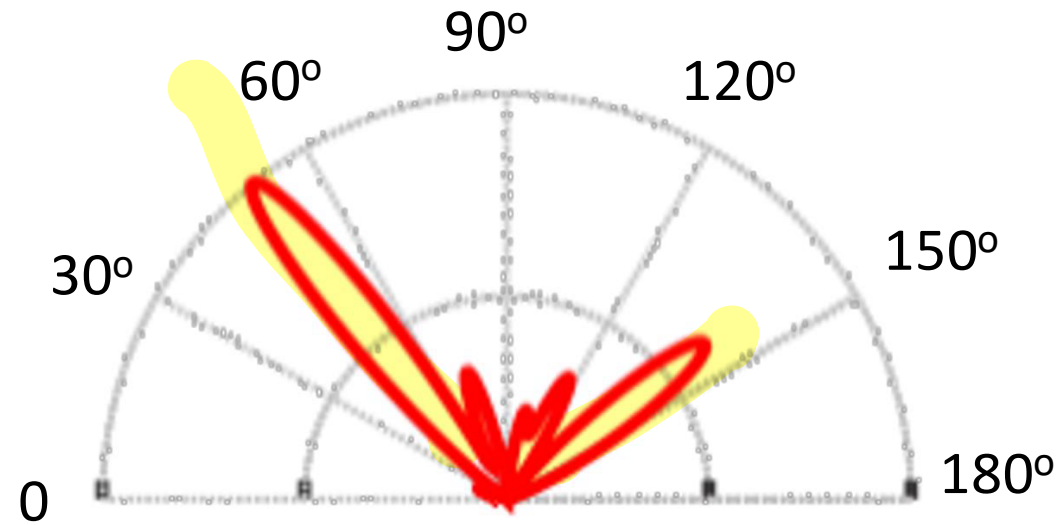
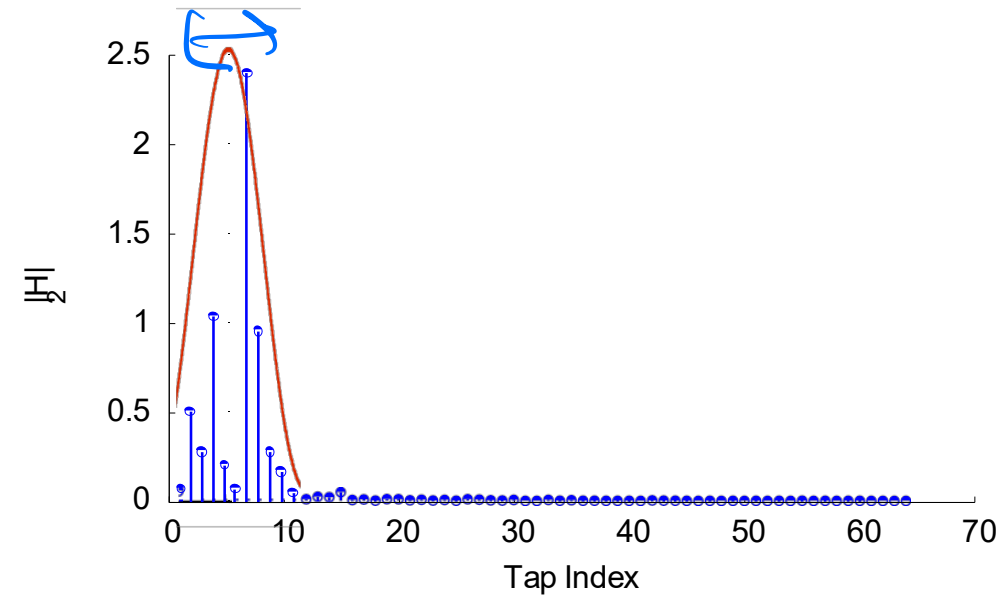
$\uparrow$
time resolution

if B is low



Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?



Multipath Profile vs Time

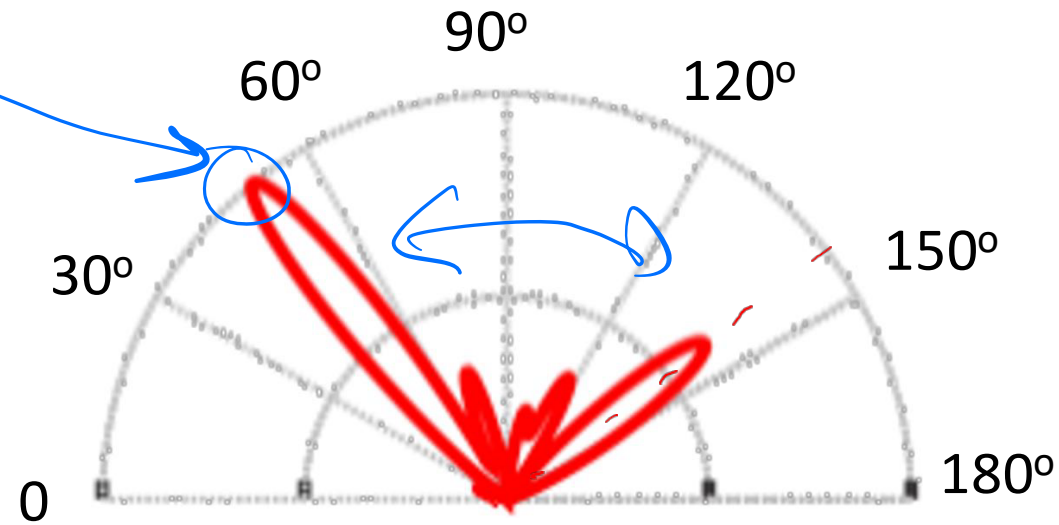
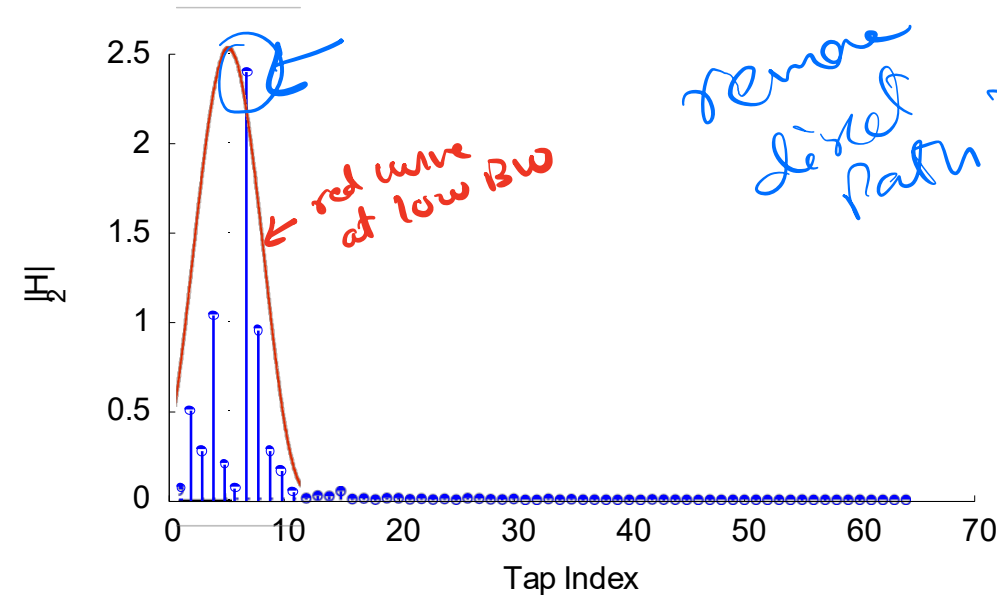
Multipath Profile vs AoA

1. Use multipath profile as a filter to separate different paths
2. Estimate time of arrival of each path
3. Find the shortest path

But How?

Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?



Multipath Profile vs Time

Multipath Profile vs AoA

1. Use multipath profile as a filter to separate different paths

$$P(\theta_1) = |y_{\theta_1}(t)|^2$$

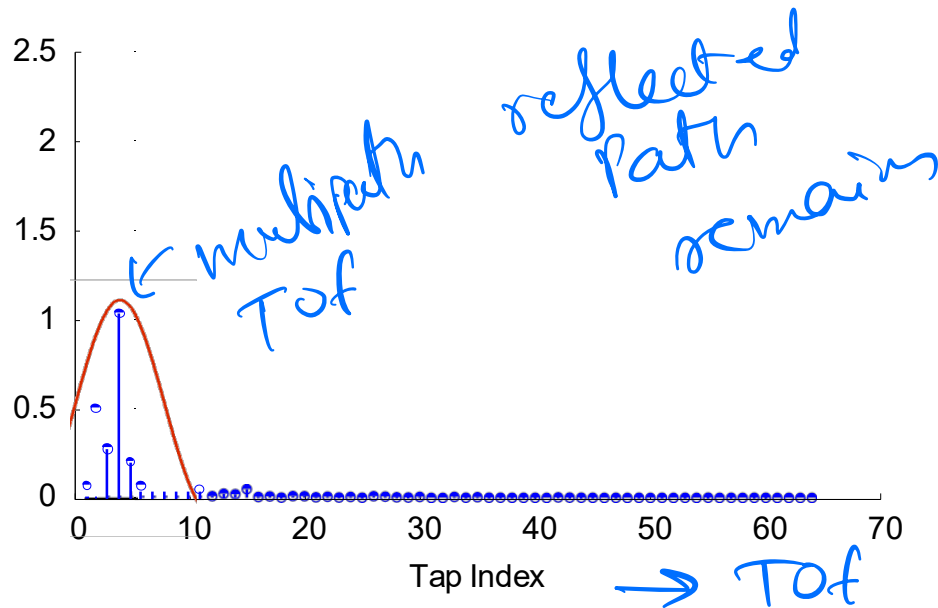
$$y_{\theta_1}(t) = \sum_{k=1}^N y(t) e^{-jk\pi \cos \theta_1}$$

$$y(t)' = y(t) - y_0(t)$$

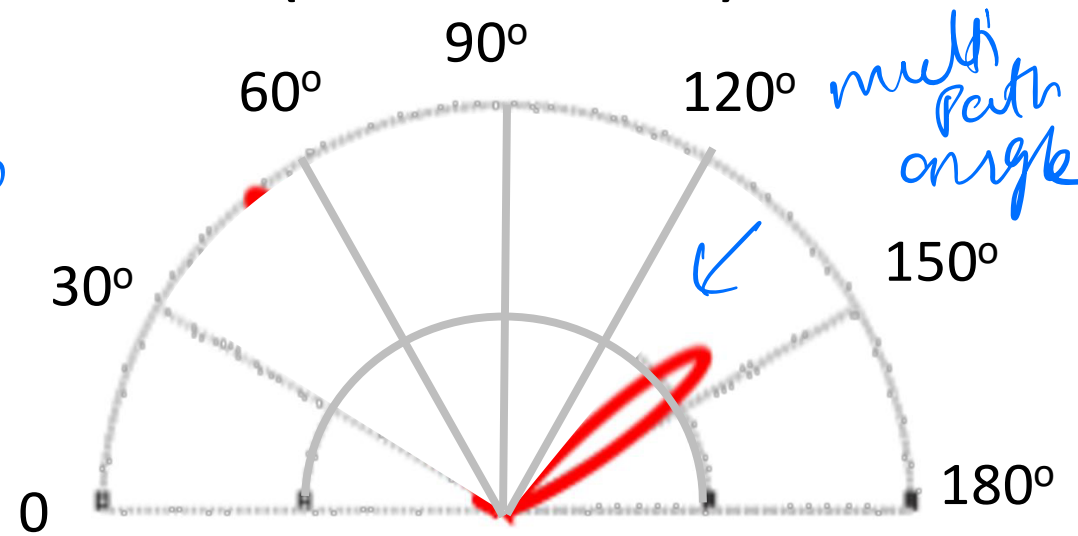
Method 3: Antenna Arrays

$y(t) \rightarrow y_{\theta_1}(t)$

Which is the Shortest Path (Direct Path)?



Multipath Profile vs Time



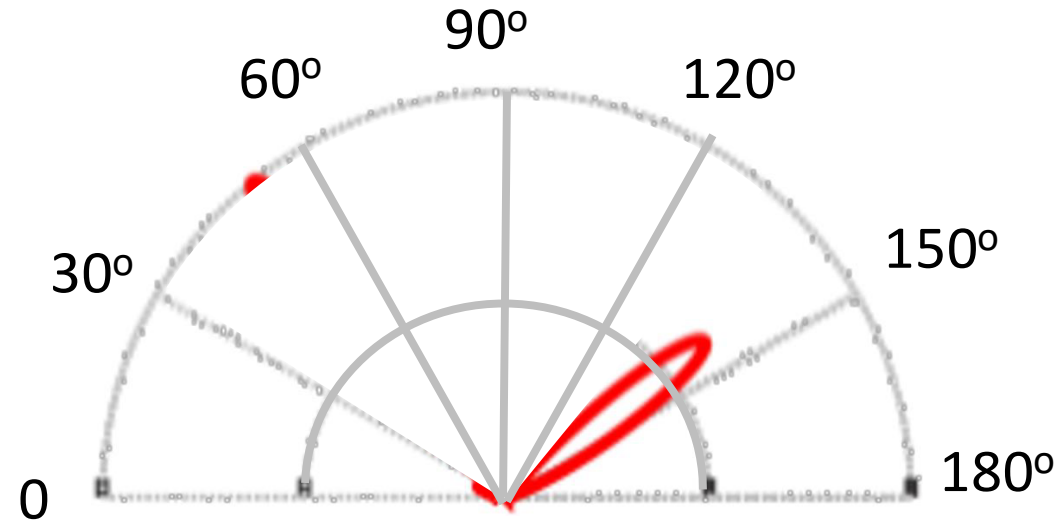
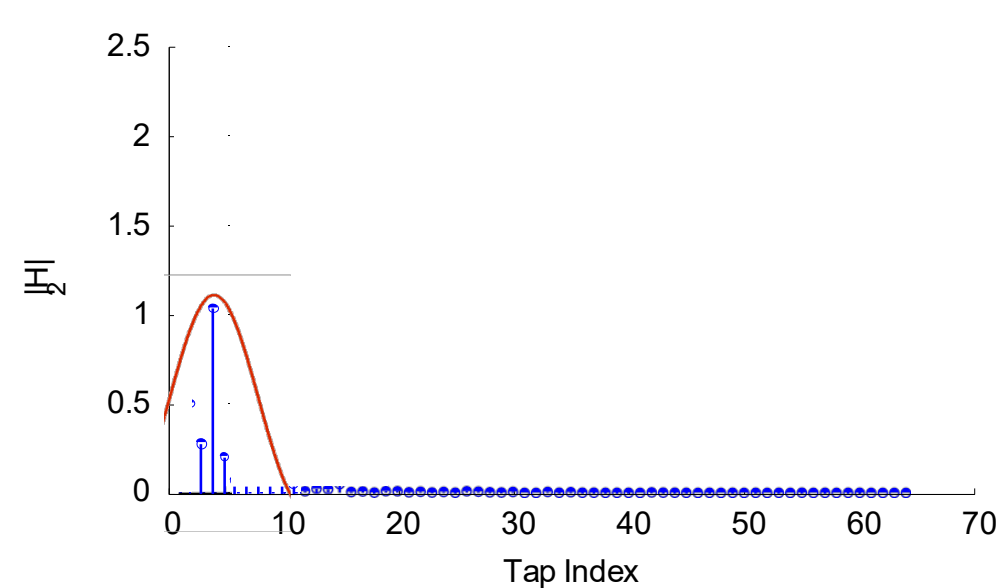
Multipath Profile vs AoA

1. Use multipath profile as a filter to separate different paths

$$y_{\theta_1}(t) = \sum_{k=1}^N y(t) e^{-jk\pi \cos \theta_1}$$

Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?



Multipath Profile vs Time

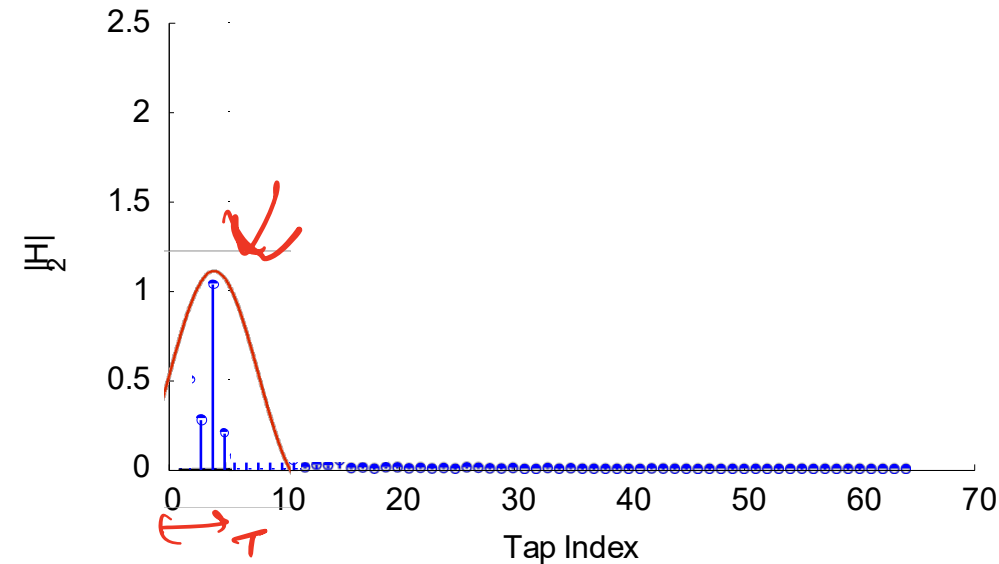
Multipath Profile vs AoA

1. Use multipath profile as a filter to separate different paths
2. Estimate time of arrival of each path

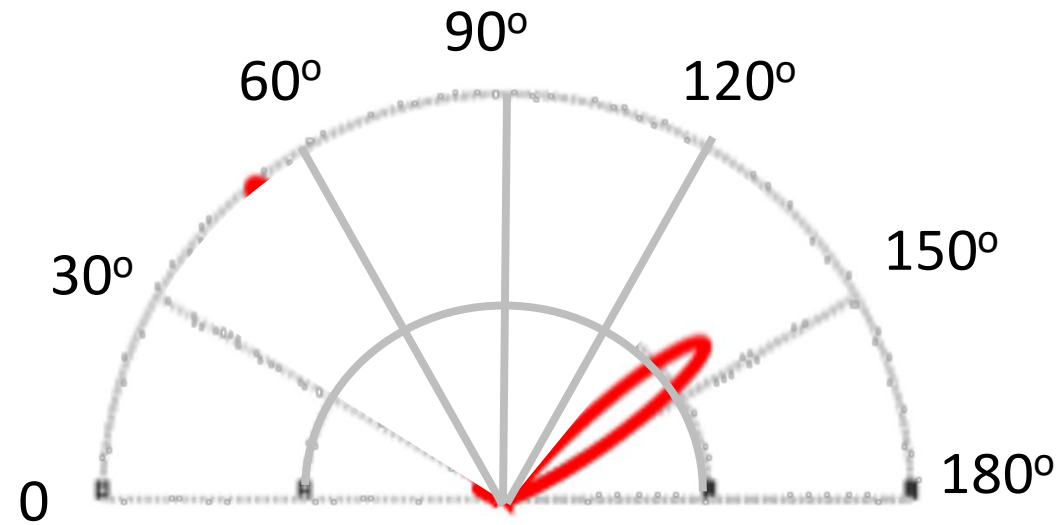
Time Resolution still not enough ➡ Use OFDM

Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?

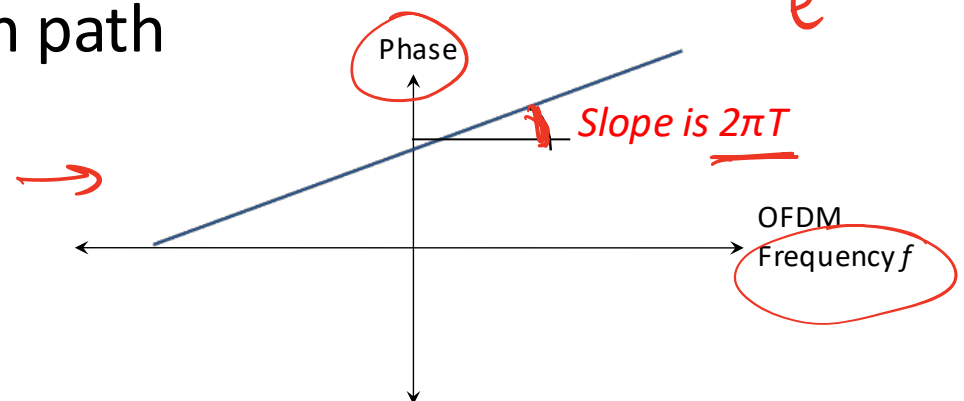
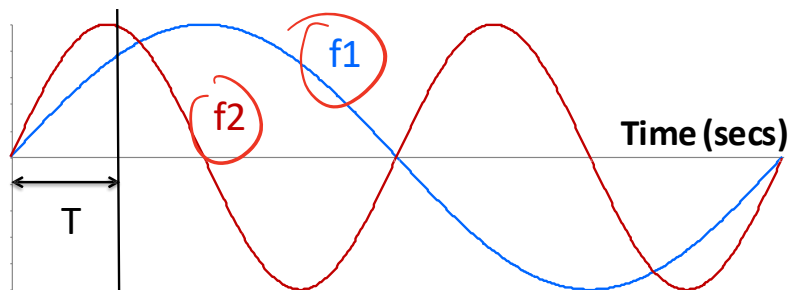


Multipath Profile vs Time



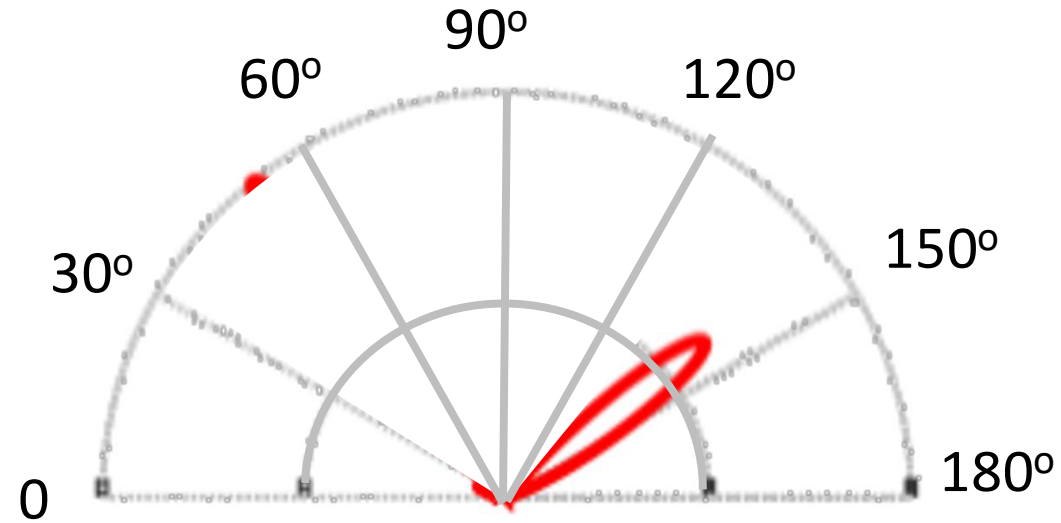
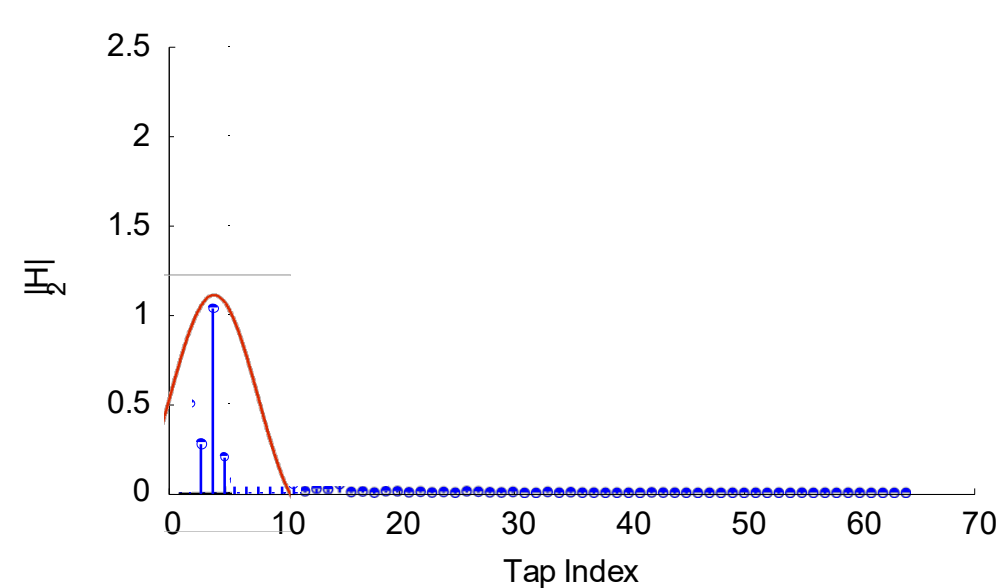
Multipath Profile vs AoA

1. Use multipath profile as a filter to separate different paths
2. Estimate time of arrival of each path



Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?



Multipath Profile vs Time

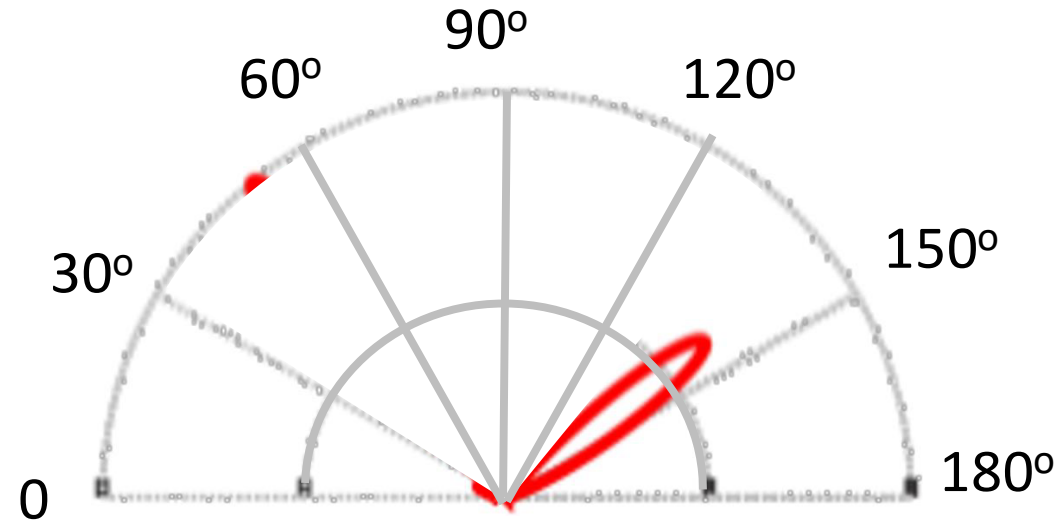
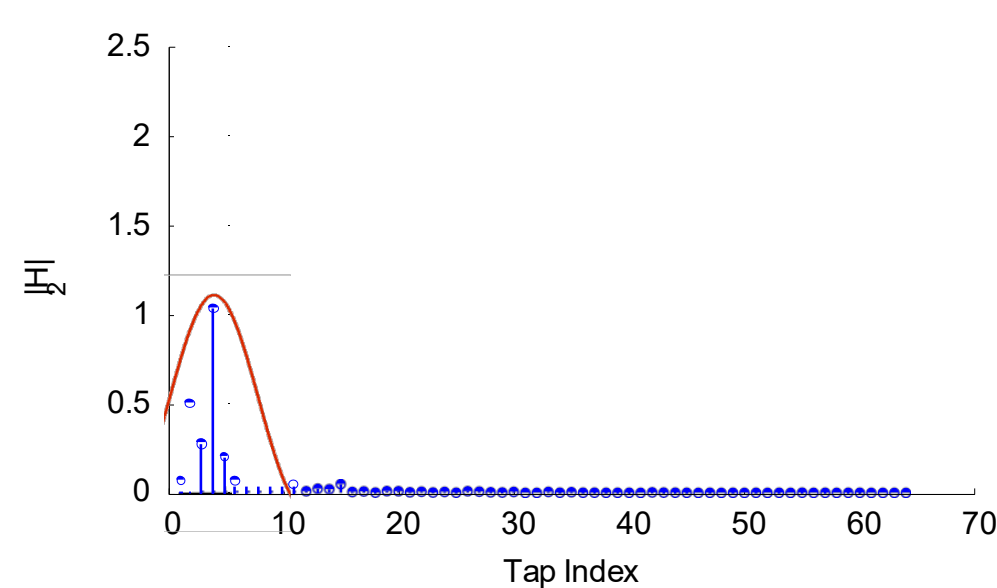
Multipath Profile vs AoA

1. Use multipath profile as a filter to separate different paths
2. Estimate time of arrival of each path
 - Use OFDM to estimate delay from slope of phase vs freq.

But this delay includes packet detection delay & processing delay, not just propagation delay!

Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?



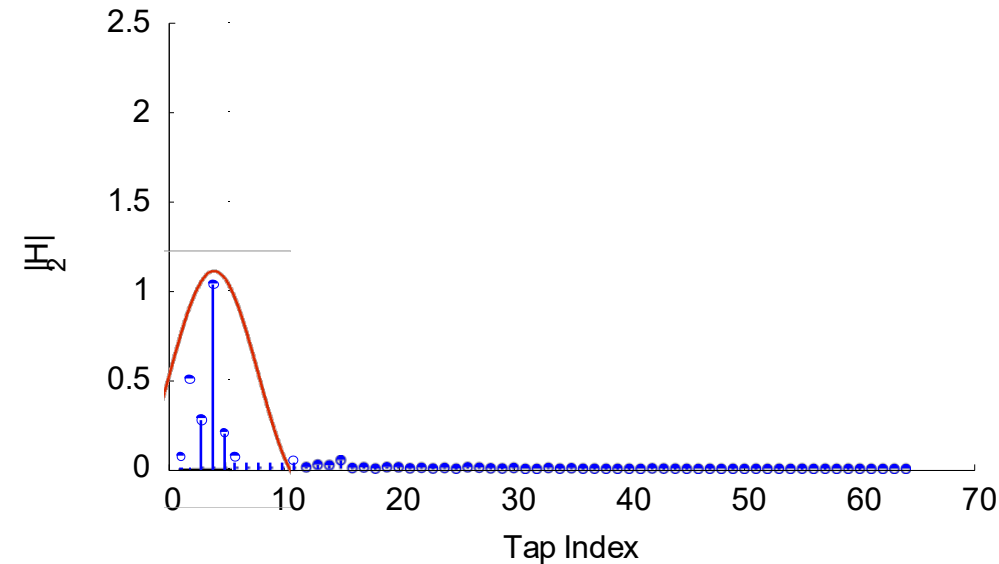
Multipath Profile vs Time

Multipath Profile vs AoA

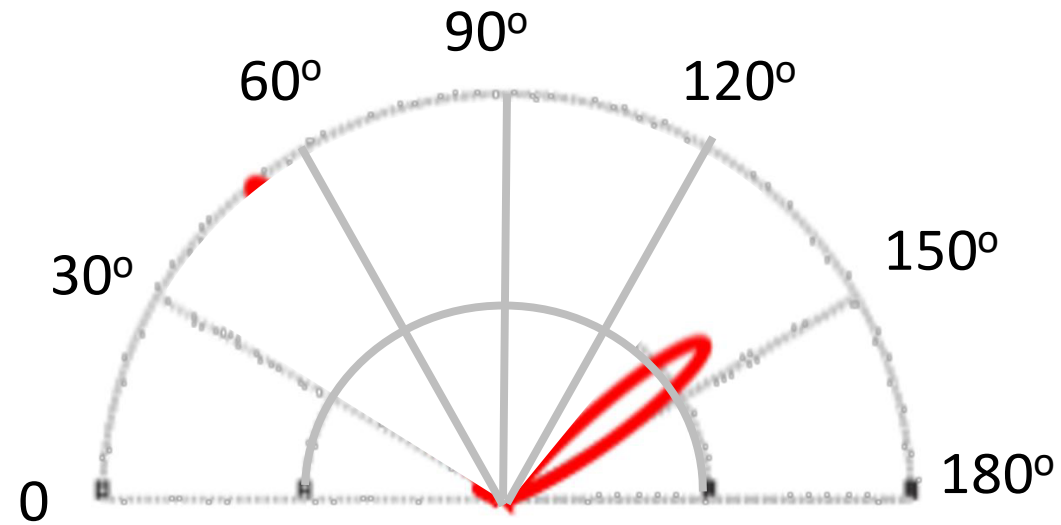
1. Use multipath profile as a filter to separate different paths
2. Estimate time of arrival of each path
 - Use OFDM to estimate delay from slope of phase vs freq.
 - Compute relative delay for different paths!
3. Compare relative delays to find the shortest path

Method 3: Antenna Arrays

Which is the Shortest Path (Direct Path)?



Multipath Profile vs Time

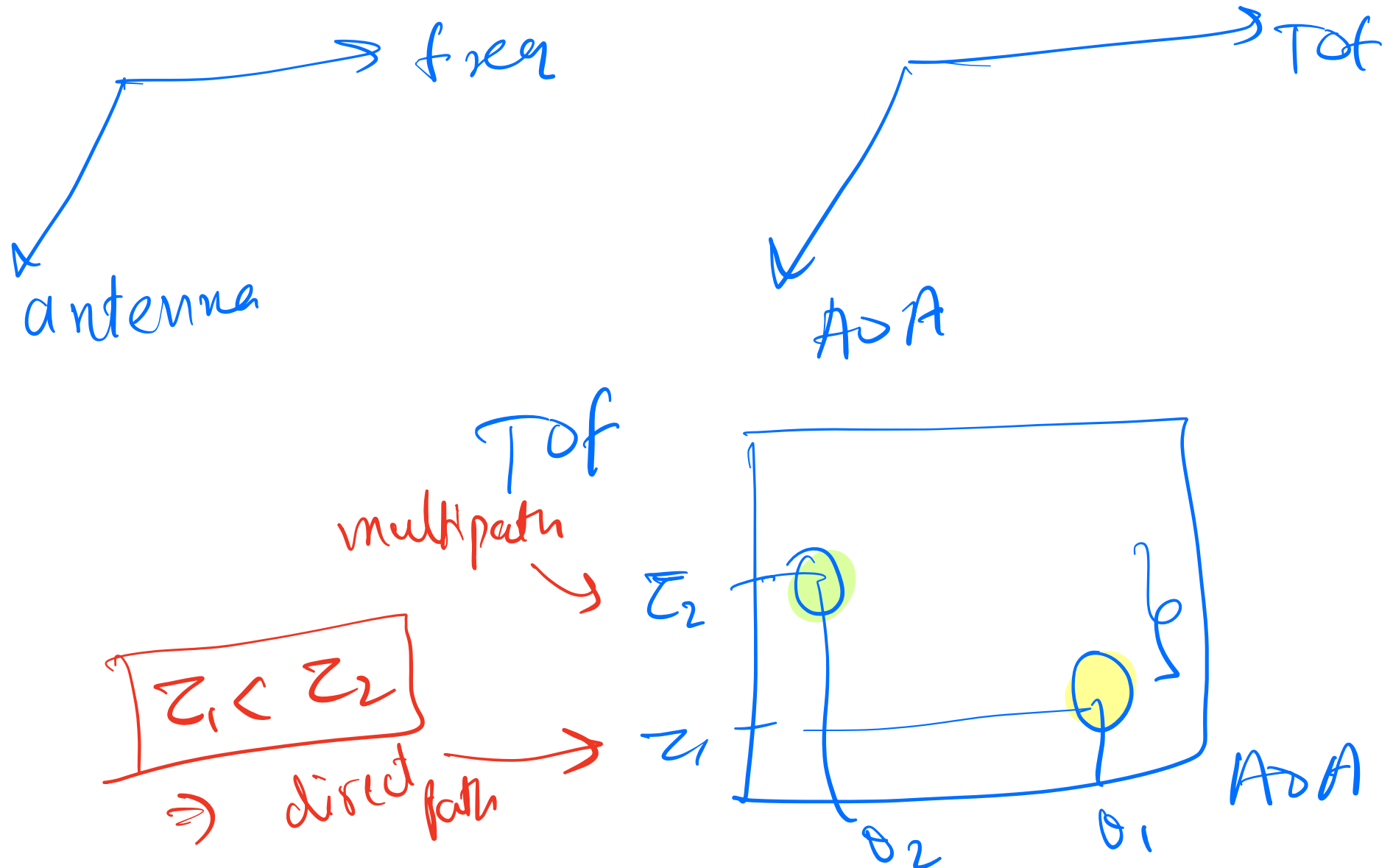


Multipath Profile vs AoA

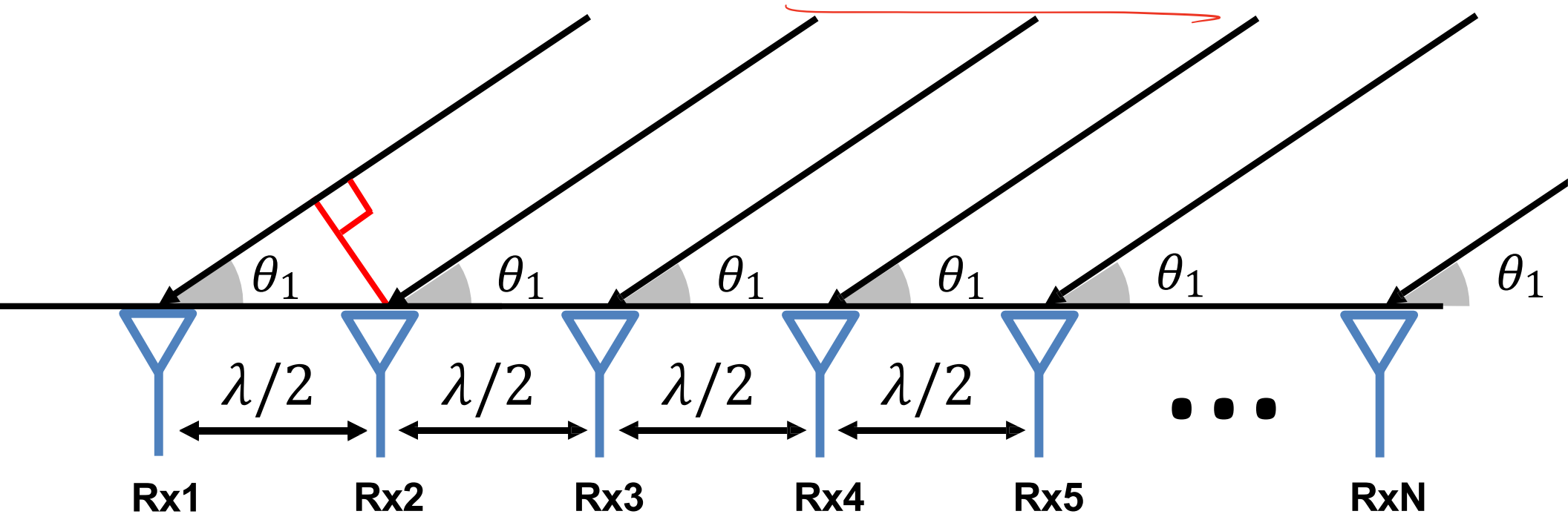
AOA - TOF Transform

Method 3: Antenna Arrays

AOA-TOF Transform



Method 3: Antenna Arrays

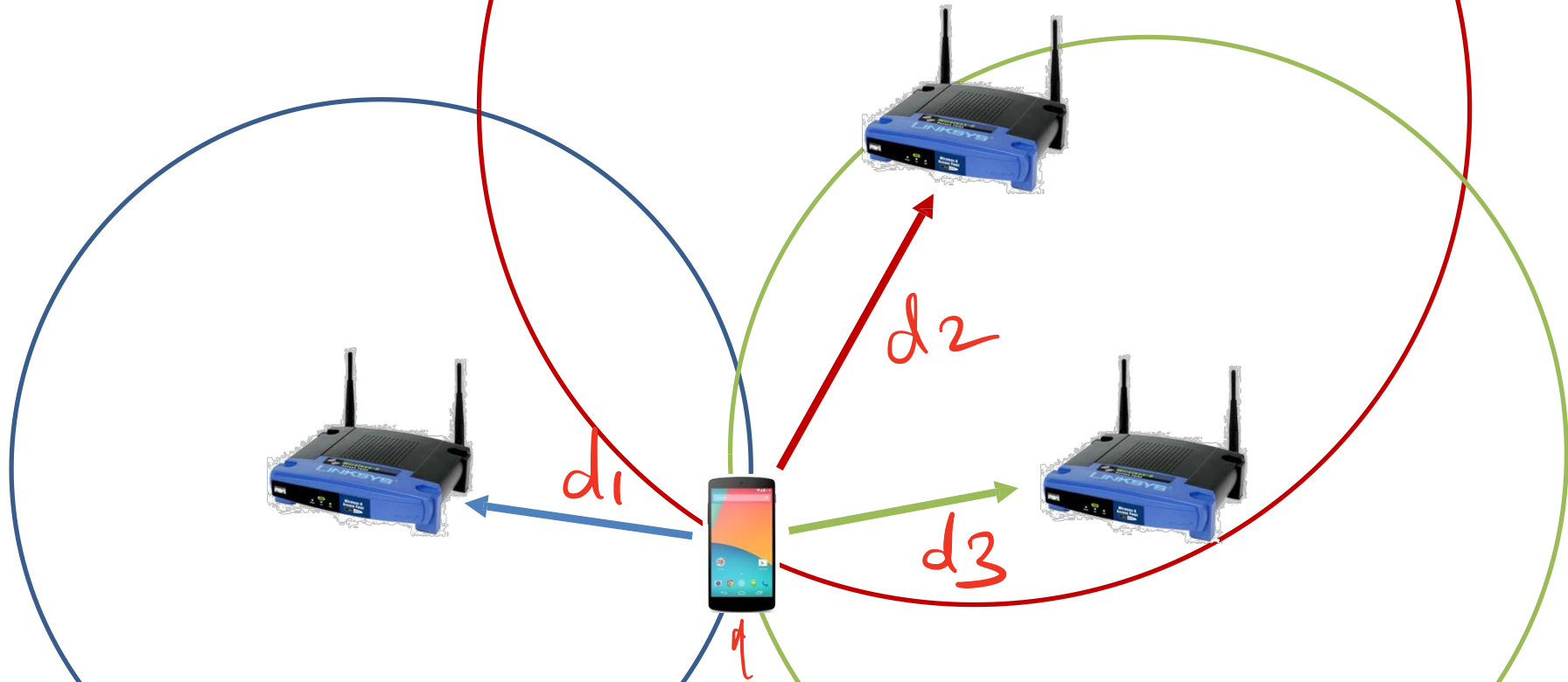


Pros: Works with multipath, No need for fingerprinting

Cons: Requires more hardware! $\Rightarrow$ more antennas $\Rightarrow$ more bandwidth
Assumes device is sufficiently far such that
wavefront is parallel far field

Method 4: ToF Based Localization

Measure Time of Flight (ToF) from device to each AP



Distance = Time of flight $\times$ speed of travel

Measure ToF $\rightarrow$ Get distance $\rightarrow$ Trilateration

Method 4: ToF Based Localization

Measure Time of Flight (ToF) from device to each AP

Challenges:

- How do you know when signal was transmitted?



Measure Round Trip

- How about packet detection delay & processing delay?
 - Use OFDM to correct for packet detection delay
 - Estimate and calibrate for processing delay

Not Practical!

*not used for
wifi*

Method 4: ToF Based Localization

Measure Time of Flight (ToF) from device to each AP

Challenges:

- Resolution limited by sampling rate (bandwidth)!

$$\Delta d = \Delta \tau \times c$$

Handwritten notes: $\frac{1}{B} = \frac{1}{40 \times 10^6}$

802.11n bandwidth = 40MHz $\Rightarrow \Delta \tau = 25ns \Rightarrow \Delta d = 12.5 m$

Handwritten notes: $\uparrow$ resolution, $\times 3 \times 10^8$, $\uparrow$ distance resolution

- Other systems than WiFi can get accurate ToF:

- UWB: Ultra-Wide Band

- FMCW: Frequency Modulated Carrier Wave

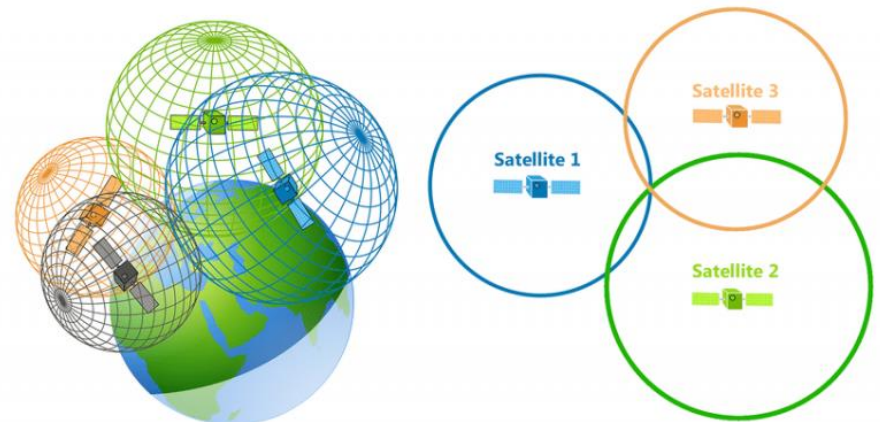
Not Supported in WiFi
(Will discuss in future lectures)

Method 4: ToF Based Localization

Measure Time of Flight (ToF) from device to each AP

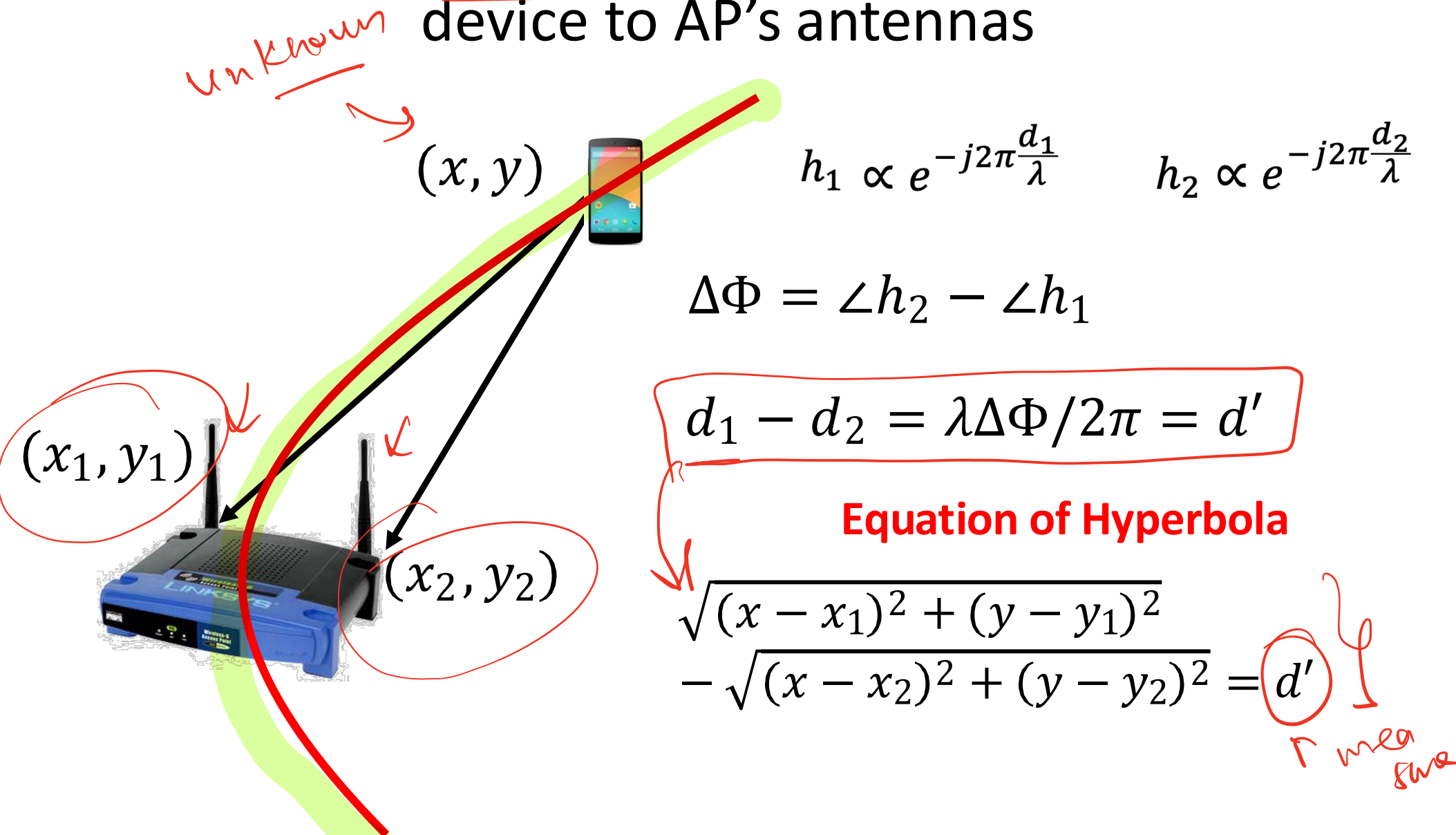
Challenges:

- How do you know when signal was transmitted?
- How GNSS Positioning work then?
- Satellite beacon sends its positions and time sent.
- Device estimates time-difference between satellite clock and device clock → Distance
- All satellites are perfectly synchronized
- Four unknown – x, y, z , clock bias → 4 equations from at least 4 satellites.



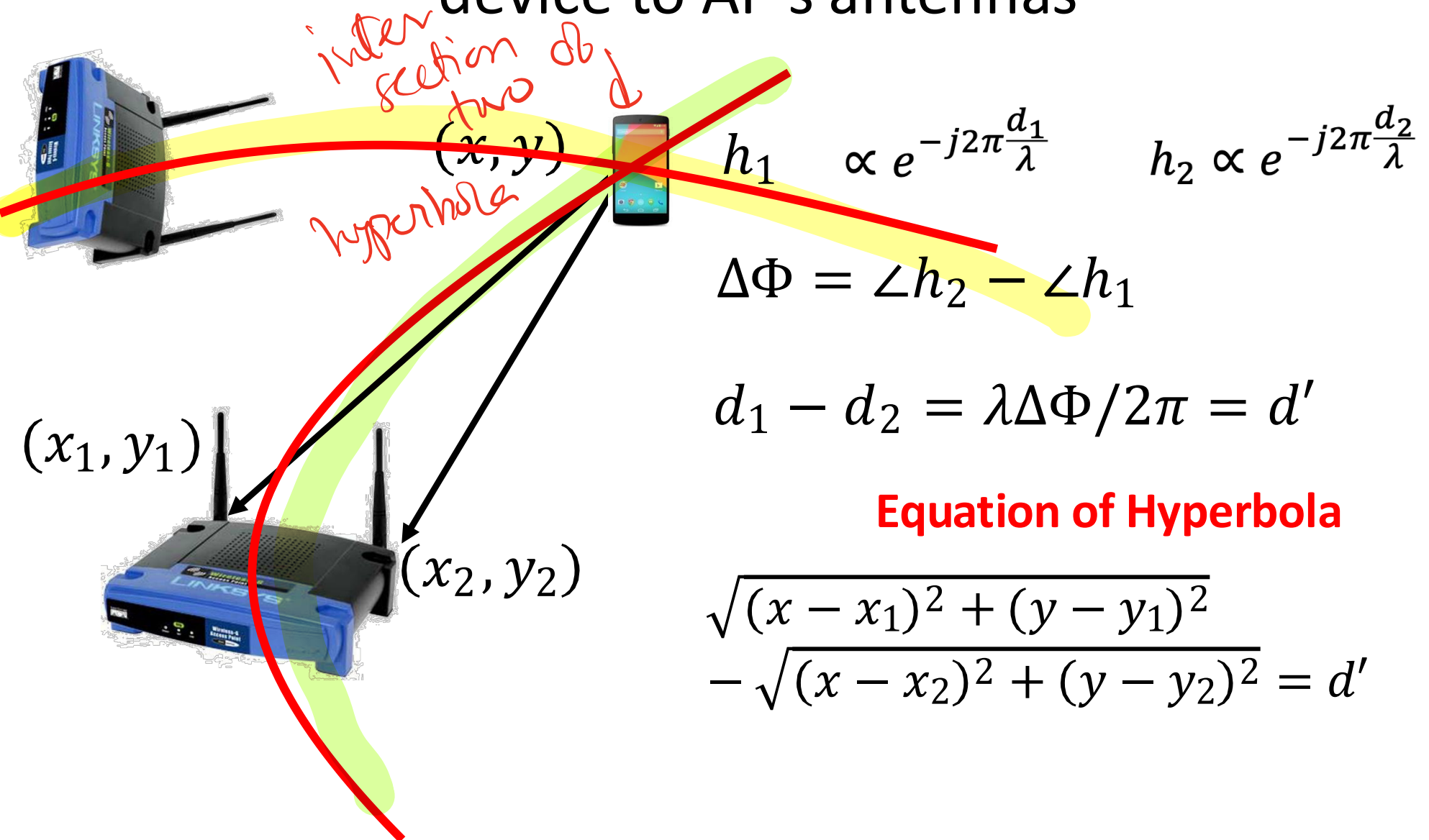
Method 5: TDoA Based Localization

Measure Time Difference of Arrival (TDoA) from device to AP's antennas

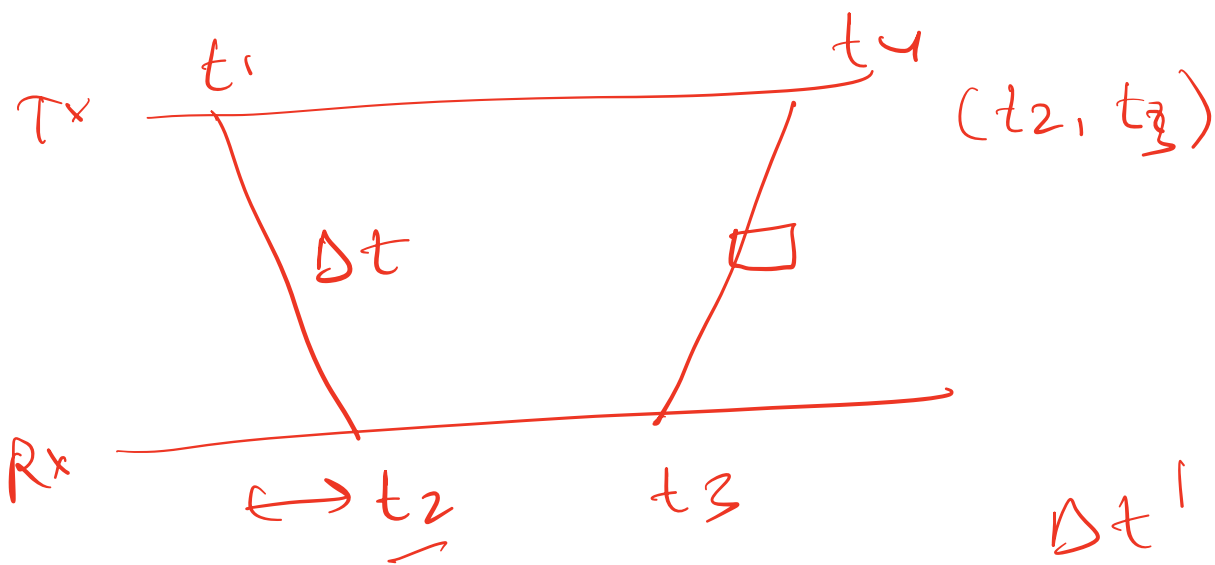


Method 5: TDoA Based Localization

Measure Time Difference of Arrival (TDoA) from device to AP's antennas



Two-way ranging (UWB)



$$t_2 = t_1 + \Delta t + t_{\text{offset}}$$

$$t_4 = t_3 + \Delta t + t_{\text{offset}}$$

$$\Delta t = \frac{(t_4 - t_1) - (t_3 - t_2)}{2}$$